2026 prelim mechanics answers

Answer pages for the 2026 prelim mechanics questions.
Whole pages are kept. A page that also holds another answer is included in full.
2026 RI has no answer scheme in the corpus, so RI is not in this file.
DHS, NYJC and VJC answers are the worked papers. EJC and HCI answers are solution booklets.

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1 An object of mass 0.050 kg is projected from the top of a 50 m tall building at an angle of 60° to the horizontal with an initial speed of 20 m s^{-1} as shown in Fig. 1.1.

A is a point in the trajectory that is at the same height as the object initially and *d* is the horizontal distance the object lands from the building.

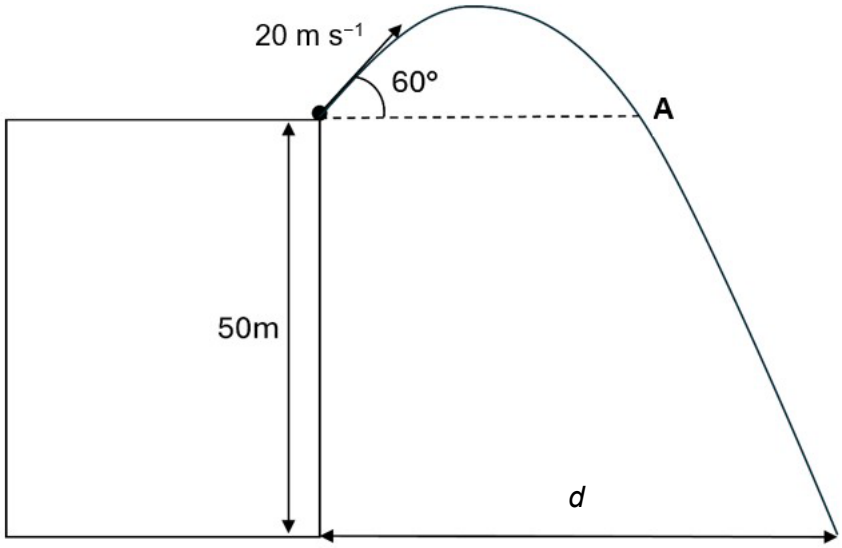


Fig. 1.1

(a) Assuming there is no resistance,

(i) Calculate the distance *d*.

taking the upward direction as positive,

vertical component of initial velocity, $u_y = 20\sin 60°$ (M1)

let t to be the time taken to reach the ground,

$s_y = u_y t + \frac{1}{2} a_y t^2$

$-50 = 20\sin 60° t - 4.905t^2$ (M1)

$t = 5.414$ s

$d = u_x t = (20\cos 60°)(5.414) = 54.1$ m (A1)

d = ... m [3]

(ii) Without calculation, state the speed of the object at **A** and explain how you arrived at your answer.

20 m s^{-1} [B1]

As the projectile rises to its maximum height, some of its kinetic energy is converted into gravitational potential energy. As it falls from the maximum height to point A (which is at the same height as the point of projection), the same amount of gravitational potential energy is converted back into kinetic energy.

Since mechanical energy is conserved (neglecting air resistance), the kinetic energy at point A is equal to the initial kinetic energy. Hence, the speed at A is the same as the initial speed. [B1]

OR

The projectile is at the same height initially and at point A, so its gravitational potential energy is the same at both positions.

As mechanical energy is conserved (neglecting air resistance), its kinetic energy must also be the same at these two positions. Therefore, its speed at A is equal to its initial speed. [B1]

..... [2]

(b) Experimentally, it is found that the same object now reaches a maximum height of 60.5 m above the ground in its trajectory.

(i) Show quantitatively that the air resistance is not negligible in this case.

neglecting the air resistance,
the maximum height s_y above the point of projection is given by:

$v_y^2 = u_y^2 + 2a_y s_y$

$0 = (20\sin 60°)^2 + 2(-9.81)s_y$

$s_y = 15.2905 \text{ m} = 15.3 \text{ m}$ (to 3 s.f.)

$\Rightarrow$ the maximum height above the building top $= 15.3$ m (M1)

factoring in air resistance,
the maximum height above the building top $= 10.5$ m

since 15.3 m > 10.5 m, (M1)
the air resistance is not negligible in the stated case

[2]

(ii) The speed of the object at its maximum height is 8.0 m s^{-1}.
Calculate the energy lost due to air resistance at maximum height.

initial kinetic energy, $KE_i = \frac{1}{2}mu^2 = 10.0$ J

kinetic energy at its maximum height, $KE_f = \frac{1}{2}mv^2 = 1.6$ J

gain in gravitational potential energy,
$\Delta GPE = mgs_y = (0.050)(9.81)(10.5) = 5.15$ J (M1)

By conservation of energy,
$KE_i = KE_f + \Delta GPE +$ energy lost due to air resistance
energy lost $= (KE_i - KE_f) - \Delta GPE = (10.0 - 1.6) - 5.15 = 3.25$ J (A1)

energy lost = ... J [2]

(c) Sketch the path of the object in Fig. 1.1 when air resistance is not negligible. [1]

[Total: 10]

2 In a collision between two bodies A and B, the force that A exerts on B varies with time in the way shown in Fig. 2.1.

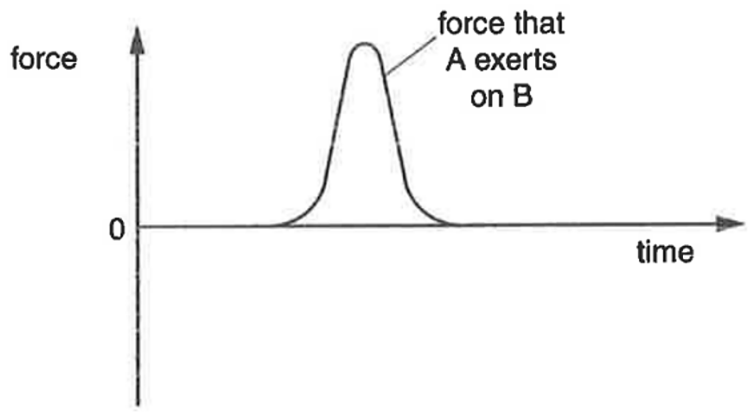
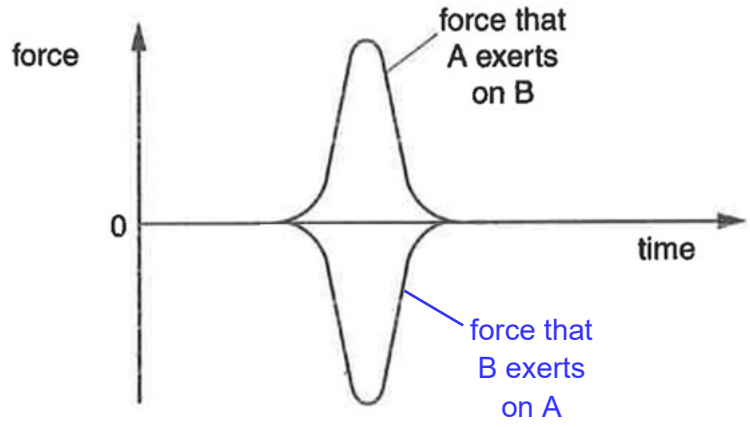


Fig. 2.1

(a) (i) Copy Fig. 2.1 and show in Fig. 2.2 a graph of the force that B exerts on A.



[B1]

Fig. 2.2 [1]

(ii) Explain your answer to **(i)**.

The forces are an action-reaction pair; therefore, they should always be equal in magnitude and opposite in direction. [B1]

……

……

…… [1]

(b) In a collision, when a truck of mass 12 000 kg runs into the front of a car of mass 1200 kg, a constant force of 72 000 N acts for 0.25 s.
Calculate the change in velocity of

(i) the car,

Since the truck runs into the front of the car (as stated), this describes a head-on collision.

impulse = $F \times \Delta t = 72000 \times 0.25 = 18000$ N s = 18000 kg m s^{-1} [B1]

Both the truck and the car experience equal and opposite impulses (Newton's Third Law) or change in momentum.

change in velocity of the car, $\Delta v = \Delta p / m_{car}$
$= (F\Delta t)/m_{car} = 18000/1200 = 15$ m s^{-1}, [B1]

in the direction of the truck's initial motion.

change in velocity = ... m s^{-1} [2]

(ii) the truck,

impulse = $F \times \Delta t = 72000 \times 0.25 = 18000$ N s = 18000 kg m s^{-1} [B1]

change in velocity of the truck, $\Delta v = \Delta p / m_{truck}$
$= (F\Delta t)/m_{truck} = 18000/12000 = 1.5$ m s^{-1}, [B1]

In the direction opposite to its own initial motion.

change in velocity = ... m s^{-1} [2]

(c) State and explain how the force on the driver could be reduced by safety features in a typical car.

Seat belts, airbags and crumple zones increase the time taken for the driver to come to rest. [B1]

For the same change in momentum, increasing the time of impact reduces the average force acting on the driver. [B1]

.....

..... [2]

[Total: 8]

2 The Earth has a mass M of 5.97×10^{24} kg. A satellite of mass m orbits the Earth in a circular geostationary orbit at a distance d above the Earth's surface.

(a) Show that the radius of the Earth, R, is approximately 6370 km.

$$g = \frac{GM}{R^2}$$

$$\Rightarrow R = \sqrt{\frac{GM}{g}} = \sqrt{\frac{\left(6.67 \times 10^{-11}\right)\left(5.97 \times 10^{24}\right)}{9.81}} \quad \text{[B1]}$$

$$= 6.37 \times 10^{6} \text{ m} = 6370 \text{ km} \quad \text{[A0]}$$

[1]

(b) By considering the orbit of the geostationary satellite, show that the distance d above the Earth's suface is 35900 km.

gravitational force acting on satellite by the Earth
provides the centripetal force on satellite:

$$\Rightarrow \frac{GMm}{r^2} = \frac{mv^2}{r} = \frac{m(r\omega)^2}{r} = \frac{m\left(r\frac{2\pi}{T}\right)^2}{r} \quad \text{[B1]}$$

$$\Rightarrow T^2 = \frac{4\pi^2 r^3}{GM} \Rightarrow r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

$$\Rightarrow r = \sqrt[3]{\frac{\left(6.67 \times 10^{-11}\right)\left(5.97 \times 10^{24}\right)(24 \times 60 \times 60)^2}{4\pi^2}} = 42226.9 \text{ km} \quad \text{[B1]}$$

$$d = 42226.9 - 6370 = 35856.9 \text{ km} = 35900 \text{ km (to 3 s.f.)} \quad \text{[A0]}$$

[2]

(c) For the satellite of mass m that is in a circular orbit of radius r around the Earth of mass M, show that its total mechanical energy E can be expressed as $E = -\frac{GMm}{2r}$.

gravitational force provides the centripetal force:

$$\Rightarrow \frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow E_k = \frac{1}{2}mv^2 = \frac{GMm}{2r} \quad \text{[B1]}$$

$$E = E_k + E_p$$

$$E = \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right) \quad \text{[B1]}$$

$$E = -\frac{GMm}{2r} \quad \text{[A0]}$$

[2]

(e) The satellite has a mass of 490 kg and carries 360 kg of lithium battery pack with energy density of 324 Wh kg^{-1}, of which 65% of its energy is converted to useful energy for orbital manoeuvres.

(i) Show that the amount of useful energy is 2.73×10^8 J.

useful energy = $360 \times 324 \times 60 \times 60 \times 0.65 = 2.73 \times 10^8$ J [A1]

[1]

(ii) Calculate the maximum orbital radius r_{max} the satellite can reach from its geostationary orbit.

$$\Delta E = E_f - E_i = -\frac{GMm}{2}\left(\frac{1}{r_{max}} - \frac{1}{r_i}\right)$$

$$2.73\times10^8 = -\frac{\left(6.67\times10^{-11}\right)\left(5.97\times10^{24}\right)(850)}{2}\left(\frac{1}{r_{max}} - \frac{1}{4.22\times10^7}\right) \quad \text{[M1]}$$

$$\frac{1}{r_{max}} - \frac{1}{4.22\times10^7} = -1.613\times10^{-9}$$

$r_{max} = 45300$ km [A1]

r_{max} = ... m [2]

[Total: 8]

4 A non-uniform L-shaped beam of weight 10 N is attached to a wall using a hinge. The longer part of the beam is 50 cm and the shorter part of the beam is 10 cm. The weight of the beam acts at a point on the beam that is at a horizonal distance of *x* away from the hinge. The beam is held at rest using a cord, such that the longer part of the beam is horizontal. *AO* is horizontal while *OC* is vertical. The tension in the cord is 5.0 N, at an angle of 60° from *OC*, as shown in Fig. 4.1.

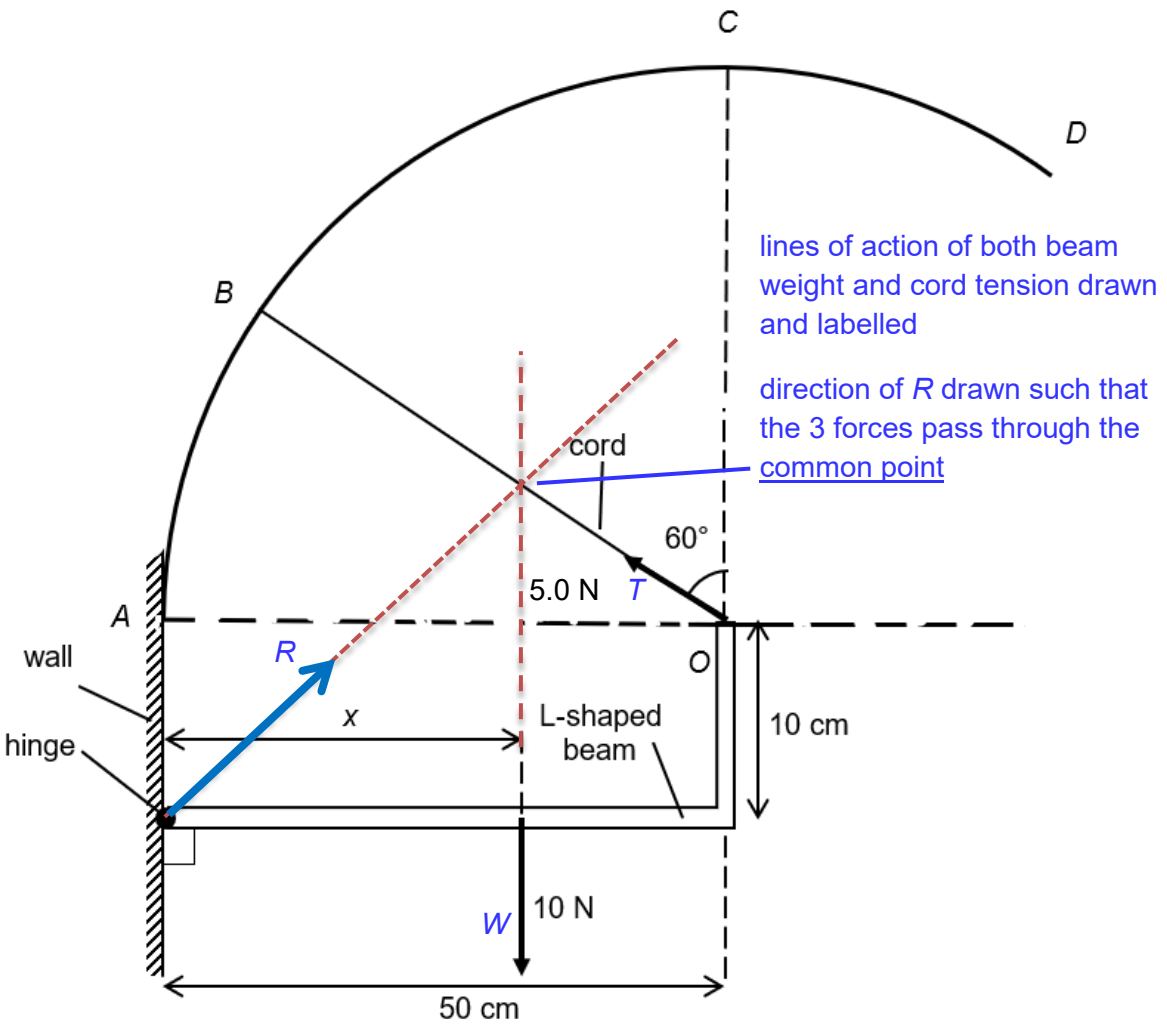


Fig. 4.1

(a) On Fig. 4.1, draw and label the contact force *R* acting on the L-shaped beam due to the hinge.

[2]

(b) By considering the forces acting on the beam, calculate the magnitude of R.

$\sum F_x = 0 \Rightarrow R_x = T_x = 5.0\sin 60^\circ = 4.33$ N

$\sum F_y = 0 \Rightarrow R_y + T_y = W \Rightarrow R_y = W - T_y = 10 - 5.0\cos 60^\circ = 7.50$ N

} [M1]

$|R| = \sqrt{R_x^2 + R_y^2} = \sqrt{4.33^2 + (-7.50)^2}$ [M1]

$= 8.66$ N

$= 8.7$ N (to 2 s.f.) [A1]

magnitude of R = ... N [3]

(c) Determine the value of x.

using hinge as pivot:
clockwise moment due to weight = anticlockwise moment due to tension

$Wx = \left[T\left(\cos 60^\circ\right)\right](0.50) + \left[T\left(\sin 60^\circ\right)\right](0.10)$

$10x = 5.0\left[0.50\left(\cos 60^\circ\right) + 0.10\left(\sin 60^\circ\right)\right]$ [M1]

$x = 0.168$ m $= 0.17$ m $= 17$ cm [A1]

x = ... cm [2]

(d) *ABCD* are different points on a circle where the centre is at point *O*. *C* is at the point vertically above point *O*. The cord moves along the arc of the circle such that the beam remains stationary.

(i) State how the magnitude of the contact force *R* would change as the cord moves from *B* to *C*.

Magnitude *R* decreases as the cord moves from *B* to *C*. [B1]

..... [1]

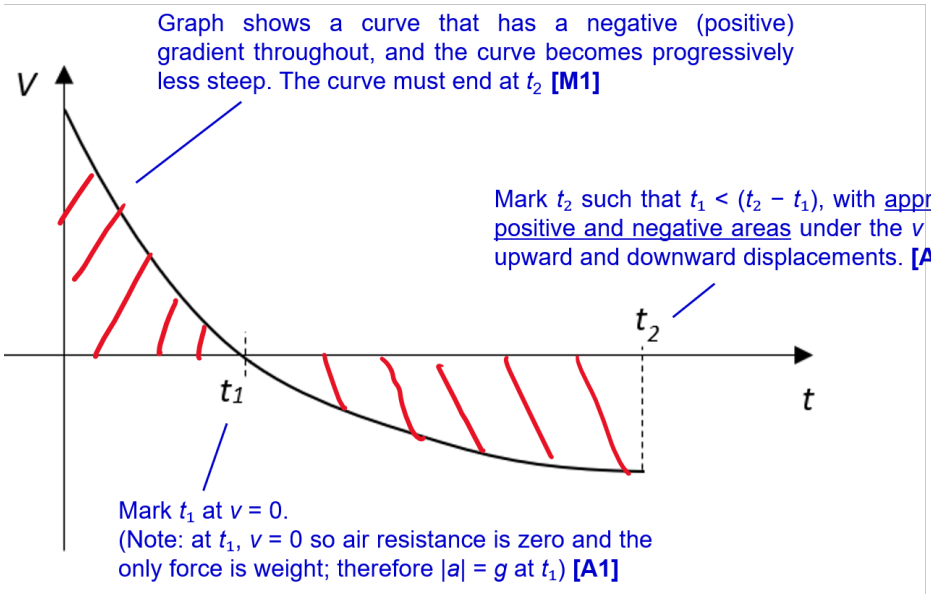
(ii) State how the direction of the contact force *R* would change as the cord moves from *C* to *D*. Mathematical calculation is not required in this part.

At *C*, *R* acts vertically upwards. As the cord moves from *C* to *D*, the direction of *R* rotates anticlockwise into the wall. [B1]

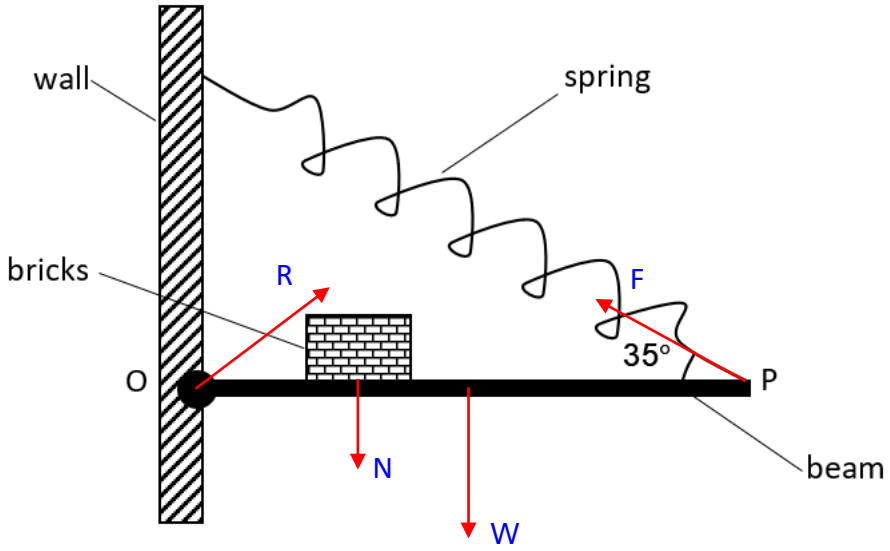
..... [1]

[Total: 9]

2026 EJC H2 Physics P2 Solutions

Qns	Answer	Marks
1(a)(1)	At t = 0, the gradient of the graph is 0. Hence the speed of the ball is 0.	A1
1(a)(2)	The gradient of the graph eventually becomes constant after the t = 0.80 s mark, indicating that the ball has reached terminal velocity. The acceleration, and thus the resultant force is zero. Since $F_{net} = mg - F_{air} = 0$, this shows that air resistance is significant and balances the weight when terminal velocity is reached.	M1 A1
1(b)	Draw a tangent to Fig. 1.1 at t = 0.40 s and determine its gradient. Speed = (2.8 ± 0.3) m s^{-1}.	M1 A1
1(c)		M1 A1 A1
	TOTAL FOR QUESTION 1	**8**
2(a)	The rate of change of momentum of a body is (directly) proportional to the resultant force acting on the body and is in the same direction as the resultant force.	B1
2(b)(i)	The sand initially has zero **horizontal** momentum but accelerates to the speed of the belt. Therefore, the belt exerts a forward horizontal force on the sand to produce this change in momentum. By Newton's third law, the sand exerts an equal and opposite horizontal force on the belt.	M1 A1
2(b)(ii)	$\langle F \rangle = \frac{m}{t}\Delta v = (0.100)(0.050)$ $= 0.0050$ N	M1 A0
2(b)(iii)1	$P = Fv = 0.0050 \times 0.050 = 2.5 \times 10^{-4}$ W	B1
2(b)(iii)2	In one second, rate at which sand gains kinetic energy, K:	

Qns	Answer	Marks
	$K = \frac{\frac{1}{2}m\Delta(v^2)}{t} = \frac{1}{2}\left(\frac{m}{t}\right)\left(v_f^{\,2} - v_i^{\,2}\right)$ $= \frac{1}{2}\left(\frac{100}{1000}\right)\left(\left(\frac{5.0}{100}\right)^2 - 0\right)$ $= 1.25 \times 10^{-4}\,\text{W}$	**B1**
2(b)(iv)	Perfectly inelastic collision. Energy transferred mechanically via work done by the motor is to the (horizontal)l kinetic store of the sand and to the internal store of the system (heating/friction).	**B1** **B1**
	TOTAL FOR QUESTION 2	**8**
3(a)	Electric potential at a point is the work done per unit charge by an external force in bringing a small positive test charge from infinity to that point.	**B1**
3(b)(i)	$E = -\frac{dV}{dr}$ $\lvert E \rvert$ = gradient of graph $= \left\lvert \frac{2.50 - 0}{(0 - 2.50) \times 10^{-2}} \right\rvert$ [M1] $= 100\,\text{N C}^{-1}$ [A1] Acceptable range: 88 <=E<= 1105□□ Direction: in positive x-direction / towards right / from left to right / → [B1]	**M1** **A1** **B1**
3(b)(ii)	The graph is symmetrical.	**B1**
3(b)(iii)	Using a point such as x = 0.20 cm where V ≈ 6.0 V: $V = \frac{Q}{4\pi\varepsilon_o x} + \frac{Q}{4\pi\varepsilon_o (0.040 - x)}$ $6.0 = \frac{Q}{4\pi\varepsilon_o}\left(\frac{1}{0.002} + \frac{1}{0.040 - 0.002}\right)$ [M1] $Q = 1.27 \times 10^{-12}\,\text{C}$ [A1] Accept also if student used other suitable points for substitution.	**M1** **A1**
	TOTAL FOR QUESTION 3	**7**
4(a)	Resistance of a circuit component is the ratio of the potential difference across the component to the current passing through it.	**B1**

Qns	Answer	Marks
1(a)	Moment of a force is the product of the force and its perpendicular distance to the pivot.	B1
1(b)(i)	$\cos 35° = \frac{6.8}{y}$ $y = \frac{6.8}{\cos 35°} = 8.3$ m extension $x = 8.3 - 7.0$ $= 1.3$ m	M1 A0
1(b)(ii)	 W: weight of beam (or gravitational force by earth on beam) N: normal contact/reaction force by bricks on beam F: force by spring on beam (or spring force / tension / elastic force) (EITHER) R: contact/reaction force by wall/hinge on beam (OR) $R_{vertical}$: vertical component of contact/reaction force by wall/hinge on beam $R_{horizontal}$: horizontal component of contact/reaction force by wall/hinge on beam (Alternatively): normal contact/reaction force by wall/hinge on beam Correct position and direction of all forces Length of arrow of W longer than N	B1 B1

Qns	Answer	Marks
1(b)(iii)	For rotational equilibrium, taking moment about the pivot O, sum of clockwise moment = sum of anticlockwise moment $N(1.8)+W(3.4)=(F\sin 35°)(6.8)$ $(120)(9.81)(1.8)+(190)(9.81)(3.4)=(F\sin 35°)(6.8)$ $F=2168$ N $F=2200$ N or $F=2170$ N $k=\frac{F}{x}=\frac{2200}{1.3}$ or $k=\frac{F}{x}=\frac{2170}{1.3}$ $=1692$ N m^{-1} $=1669$ N m^{-1} $=1700$ N m^{-1} (2 s.f.) $=1700$ N m^{-1} (2 s.f.) Accept both 2 s.f. and 3 s.f. **Note: accept 1670 N m^{-1}, 1690 N m^{-1} and 1700 N m^{-1}.**	**M1** **A1**
1(b)(iv)	$E_P=\frac{1}{2}kx^2$ $=\frac{1}{2}(1700)(1.3)^2$ $=1437$ $=1400$ J (2 s.f.) Note: Can also use $E_P=\frac{1}{2}Fx$ or $E_P=\frac{1}{2}\left(\frac{F^2}{k}\right)$ Accept both 2 s.f. and 3 s.f. Allow ECF. **Note: accept 1400 J, 1410 J, 1430 J and 1440 J**	 **C1** **A1**
1(b)(v)	As the pile of bricks is moved towards the pivot O, the clockwise moment about the pivot (due to the bricks) decreases (as the perpendicular distance from pivot to line of action of the force decreases). (The total clockwise moment decreases). A lower anti-clockwise moment is needed to maintain equilibrium. Therefore, the force exerted by the string (on the beam) decreases.	**B1** **B1**

Qns	Answer	Marks
2**(a)**	g and r are constants so a is proportional to x the negative sign shows that a and x are in opposite directions/ a is always directed towards the equilibrium	**B1** **B1**
2(b)	since $a = -\frac{gx}{r}$ and $a = -\omega^2 x$ so $\omega^2 = \frac{g}{r}$ $\omega^2 = \frac{9.81}{0.28}$ $\omega = 5.9$ rad s^{-1} *(5.92)*	**M1** **A1**
2(c)(i)	Negative sine graph max a = $\omega^2 x_o$ = 5.92 (0.05) = 1.7 m s^{-2}	**B1** **B1**
(c)(ii)	since this is a light damping oscillation, period remain the same as before and remains constant over time Amplitude consistently decreases and first peak must be less than 5.0cm	**B1** **B1**
(c)(iii)	Loss in energy $= \frac{1}{2} m\omega^2 x_{oi}{}^2 - \frac{1}{2} m\omega^2 x_{of}{}^2$ $= \frac{1}{2} \times 0.090 \times 5.9^2 \times 0.050^2 - \frac{1}{2} \times 0.090 \times 5.9^2 \times 0.025^2$ M1 $= 0.294 = 0.29$ J A1	**M1** **A1**

Qns	Answer	Marks
3(a)(i)	$Q_{\text{water at 5.0°C to water at 0°C}} = m_{\text{water}} c_{\text{water}} \Delta T$ $= 220 \times 4.18 \times (5.0 - 0)$ $= 4598$ J $= 4600$ J (2 s.f.)	**B1**
3(a)(ii)	$Q_{\text{water at 0°C to ice at 0°C}} = m_{\text{water}} L_{\text{f}}$ $= 220 \times 334$ $= 73480$ J $= 73500$ J (3 s.f.)	**B1**

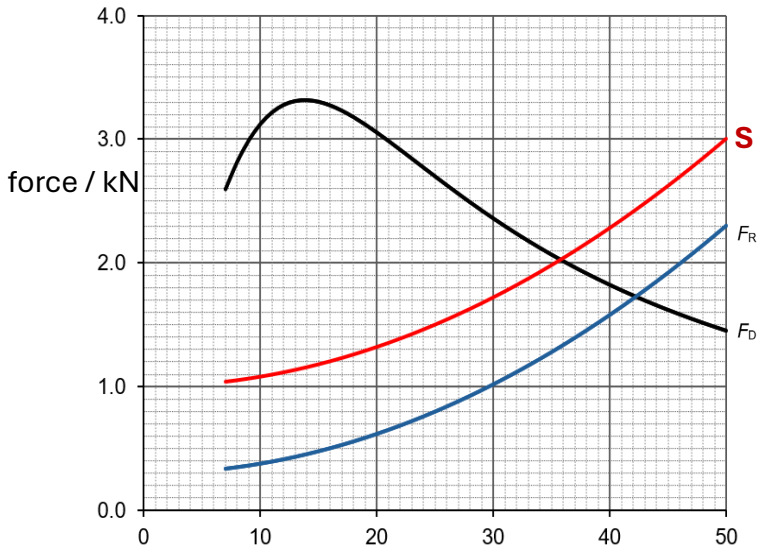
Qns	Answer	Marks
4(a)(iii)	Intensity $\propto$ Amplitude^2 Intensity = $\frac{\text{Power}}{4\pi r^2} \propto \frac{1}{r^2}$ Hence $A \propto \frac{1}{r}$ $\frac{A_{3.5}}{A_{2.5}} = \frac{r_{2.5}}{r_{3.5}}$ $\frac{A_{3.5}}{A_o} = \frac{2.5}{3.5}$ [M1] $A_{3.5} = 0.71 A_o$ [A1]	
4(b)(i)	Oscillations/ vibrations occur in a single direction (in a plane) that is perpendicular to the direction of transfer of energy/propagation.	**B1**
4(b)(ii)	Apply Malus' Law, $I_1 = I_0(\cos 40°)^2$ $I_2 = I_0(\cos 40°)^2(\cos 80°)^2$ **M1** $= 0.018 Io$ (2 s.f.) **A1**	

Qns	Answer	Marks
5(a)	When an object freefalls, the **ONLY** force acting on the object is its weight (= *mg*). **M1** Hence, the freefall acceleration $a = \frac{F}{m} = \frac{mg}{m} = g$ is equal to the gravitational field strength *g*. **A1** OR By definition of gravitational field strength, $g = F_{\text{graviational}}/m$ and from Newton's 2nd Law, for a freely falling body, the **ONLY** force acting is gravitational force hence $F_{\text{net}} = mg = ma$. Therefore, $a = g$.	
5(b)	$g = \frac{GM}{r^2}$ $M = \frac{gr^2}{G} = \frac{9.81 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}} = 5.9679 \times 10^{24} = 5.97 \times 10^{24}$	**B1**

Qns	Answer	Marks
5(c)	$\phi = -\frac{GM}{r} = -\frac{6.67\times10^{-11}\times5.97\times10^{24}}{6.37\times10^{6}} = -6.25\times10^{7}\ \text{J kg}^{-1}$	B1
5(d)	The gravitational potential is taken to be zero at infinity. Since the gravitational force is attractive, the work done by the external force to bring the test mass from infinity to a point is negative. Hence, the gravitational potential is negative.	B1 B1 A0

Qns	Answer	Marks
6(a)	The principle of superposition states that when two or more waves meet at a point, the resultant displacement is the sum of the individual displacements of each wave at that point. **Note:** Do **NOT** accept if students mention the "amplitude add up". The correct term is displacement.	B1
6(b)	Incident and reflected waves are of the same type, frequency and wavelength, and have the same speed, approximately the same amplitude and travel in opposite direction along the same tube. Incident wave from the loudspeaker reflects at the end / right side of the glass tube such that they superpose to form a stationary wave. **Note:** Do not accept if students write meet / overlap without superpose. The word interfere is not accepted as well.	B1 B1 B1
6(c)(i)	Amplitude of stationary wave: 0.06 + 0.06 = 0.12 mm	B1

Paper 2

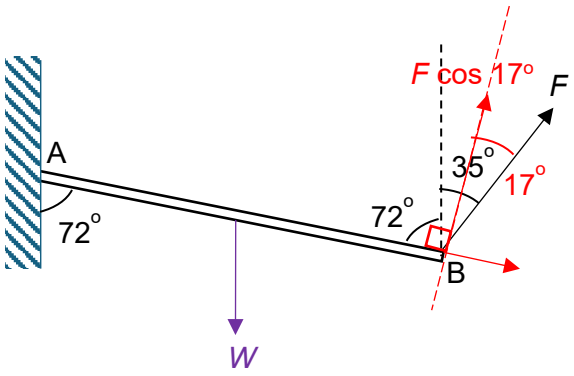
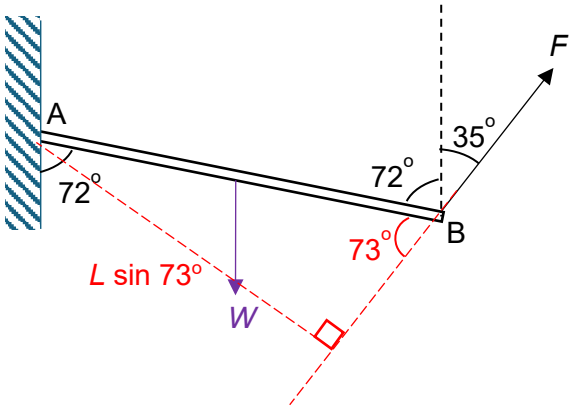
Question 1		
(a)(i)	12 to 14 m s^{-1} (The maximum acceleration occurs when the net force on the car = $F_D - F_R$ is largest.)	B1
(a)(ii)	Maximum resultant force on car, F_{net} = 3.30 kN – 0.40 kN (allow tolerance!) Maximum acceleration of the car = $\frac{(3.30-0.40)\times 10^3}{1200} = 2.42$ m s^{-2} (Allow ecf from 1(a) as long as F_{net} is > 0)	M1 A1
(a)(iii)	Rate of increase of the kinetic energy = F_{net} v = (3.30 – 0.40) × 10^3 × 12 = 3.48 x 10^4 W	C1 A1
(b)(i)	When going upslope, F_D needs to overcome both the F_R as well as the <u>component of the weight parallel to the slope.</u> Hence, the condition for <u>zero net force</u> occurs when F_D is larger and that occurs at a lower speed.	B1 B1
(b)(ii)	Correct shape – F_R shifted up by a constant value.  (The total force opposing the motion is now F_R + *mg* sin θ)	B1

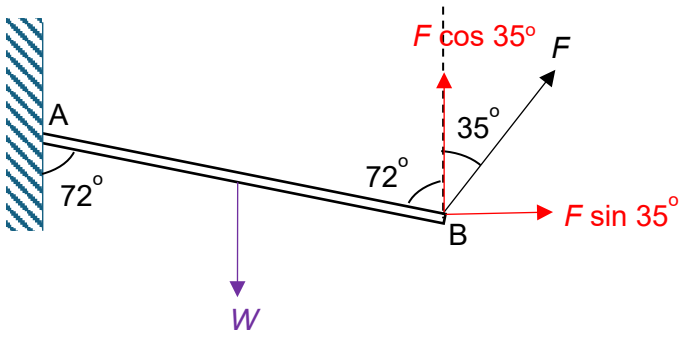
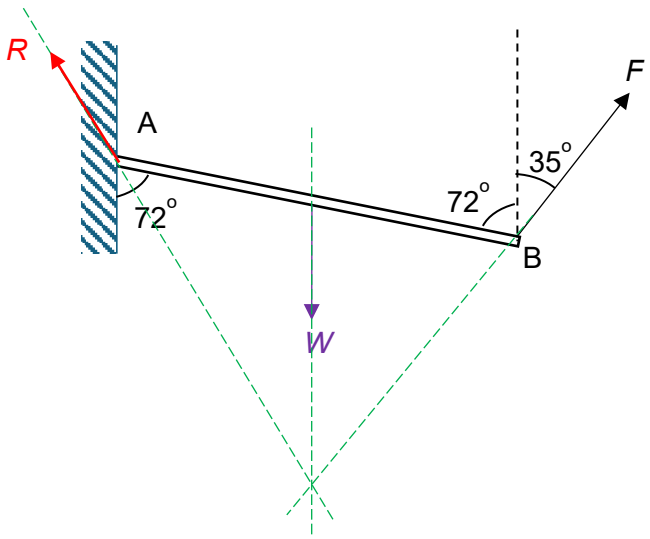
Question 2		
(a)(i)	Vertical velocity = horizonal velocity when the resultant velocity is 45° to the horizontal. $v_y = v_x = 25\cos(53°) = 15.045 = 15 \text{ m s}^{-1}$	**A1**
(a)(ii)	Taking upward as positive, $v_y = u_y + a_y t$ $15 = 25\sin(53°) + (-9.81)t$ t = 0.50 s	**A1**
(b)	$KE_{8.0} = KE_{ground} - GPE_{8.0}$ $KE_{8.0} = \frac{1}{2}(0.20)(25)^2 - (0.20)(9.81)(8.0)$ $= 47 \text{ J}$ Alternative: Accept Kinematics method.	**M1** **A1**
(c)	 **B1** – straight line with decreasing KE **B1** – non-zero KE at top of trajectory	

Question 9		
(a)	Stage 2: ${}_1^2\mathrm{H} + {}_1^1\mathrm{H} \rightarrow {}_2^3\mathrm{He} + \gamma$ Stage 3: ${}_2^3\mathrm{He} + {}_2^3\mathrm{He} \rightarrow {}_2^4\mathrm{He} + 2{}_1^1\mathrm{H}$ Accept ${}_1^1p$ for proton.	**A1** **A1**
(b)(i)	$E_K = \frac{3}{2}kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times (1.5 \times 10^7 + 273.15) = 3.105 \times 10^{-16} = 3.1 \times 10^{-16}$ J Student need to either show that they convert the temperature to kelvin or explain why they didn't convert to get the marks	**A1**
(ii)	Applying conservation of energy to determine the minimum distance the 2 protons can approach each other: Loss in kinetic energies of 2 protons = Gain in electric potential energy of protons $2E_K - 0 = \frac{e^2}{4\pi\varepsilon_0 x}$ $x = \frac{e^2}{8\pi\varepsilon_0 E_K}$ $= \frac{\left(1.60 \times 10^{-19}\right)^2}{8\pi\varepsilon_0 \left(3.1 \times 10^{-16}\right)}$ $= 3.7 \times 10^{-13}$ m Since d is larger than 10^{-14} m, the protons are not close enough to fuse.	**C1** **A1** **A1**
(c)(i)	The wave function represents the quantum state of a particle, providing all its information such as its position, momentum, etc.	**A1**
(c)(ii)1.	Since the amplitude of wave function after the barrier is non-zero, the probability density function, given by the square of the wave function, also non-zero, implies there is a possibility of finding the particle in this region.	**B1** **B1**
(c)(ii)2.	The wavelength of the wavefunction remains the same, from de Broglie's relationship, the momentum and hence energy of the particle remains the same after passing through the barrier.	**A1** **B1**

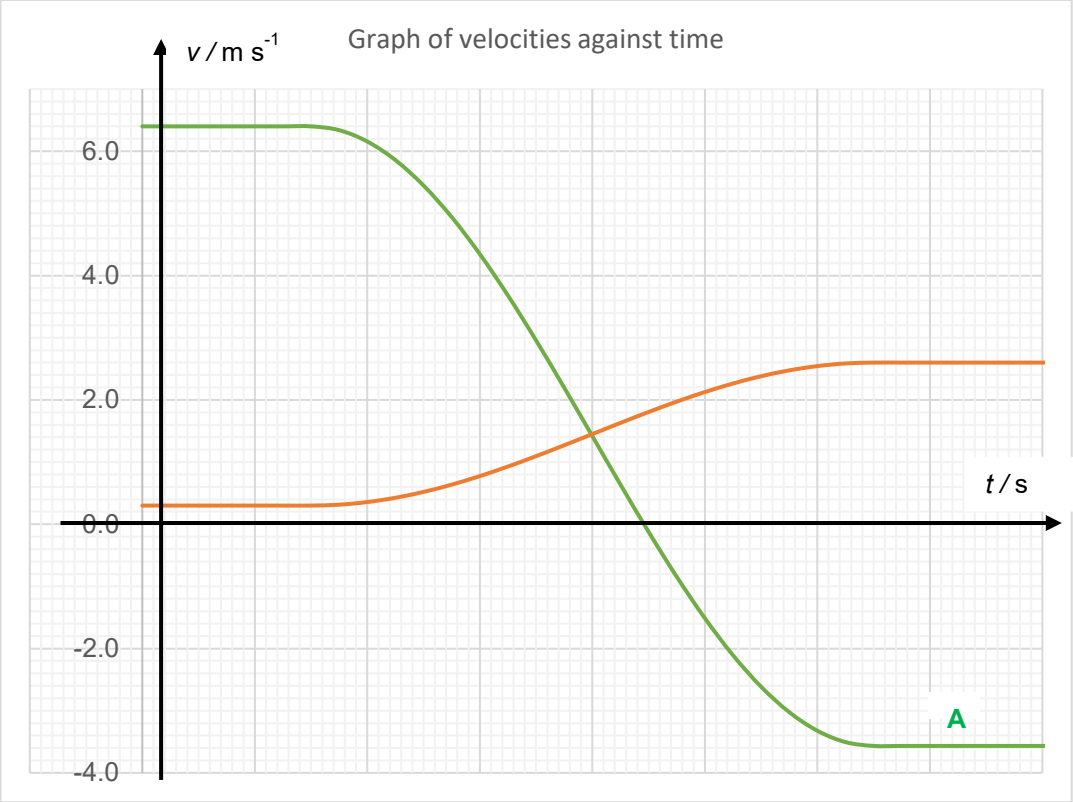
(d)(i)	Along the horizontal axis: Magnetic force provides the centripetal force $Bqv_x = mr\omega^2$ $Bqv_x = mv_x\omega$ $\frac{2\pi}{T} = \frac{Be}{m}$ $T = \frac{2\pi \times 9.11\times10^{-31}}{40\times10^{-6}\times1.60\times10^{-19}}$ $= 8.94\times10^{-7}$ s Along the vertical axis: $v_y = 5.0\times10^5 \cos 25^o$ $d = v_y T = (5.0\times10^5 \cos 25^o)\times 8.94\times10^{-7} = 0.41$ m	**B1** **B1** **B1** **A1**
(d)(ii)	Change: Opposite direction. Explain: Proton is positive. Change : Radius will increase. Explain: Proton is more massive. Any other reasonable answer.	**B1** **B1** **B1** **B1**
(d)(iii)	Charged particles in the solar transfer energy to the atoms (or electrons in the atoms) during collisions, exciting them to higher energy levels. When excited atoms/electrons in atom de-excite, photons are emitted.	**A1** **A1**
(e)(i)	Intensity $= \frac{P_{sun}}{4\pi R^2} = \sigma T^4$ $T = \left(\frac{4.0\times10^{26}}{4\pi\times(1.4\times10^{10})^2\times5.7\times10^{-8}}\right)^{1/4}$ $= 1299 = 1300$ K	**B1** **C1** **A1**
(e)(ii)	Help forecast of solar flares, changes in solar wind as modern world depend on devices are sensitive to electromagnetic variations (eg navigation systems, high voltage power grid)	**A1**

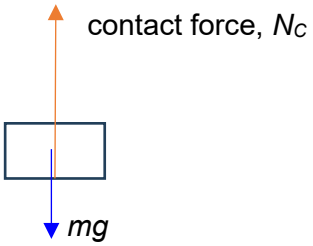
Paper 3

Question 1		
(a)	Let the length AB be L.  72°+35°-90°=17° Taking moments about A: Anti-clockwise moment due to component of F perpendicular to AB = Clockwise moment due to weight $F\cos 17^{o}\left(L\right)=W\left(0.5L\sin 72^{o}\right)$ $F=5.967=6.0$ N	**B1** **A0**
	<u>Alternative:</u>  Taking moments about A: Anti-clockwise moment due to F = Clockwise moment due to weight $F\left(L\sin 73^{o}\right)=W\left(0.5L\sin 72^{o}\right)$ $F=5.967=6.0$ N	

	Alternative:  Taking moments about A: Anti-clockwise moment due to F cos 35° + anti-clockwise moment due to F sin 35° = Clockwise moment due to weight $F\cos 35^o\left(L\sin 72^o\right)+F\sin 35^o\left(L\cos 72^o\right)=W\left(0.5L\sin 72^o\right)$ $F=5.967=6.0$ N	
(b)(i)	 Note to marker: Ignore length of arrow, working lines are expected	**A1**
(b)(ii)	For equilibrium, $\sum F_x=0$ $R_x=F_x=6.0\sin 35^o$ $\sum F_y=0$ $R_y+F_y=W$ $R_y=12-6.0\cos 35^o$ $R=\sqrt{R_x^2+R_y^2}=7.89=7.9$ N Note to marker: Substituting unrounded F gives the same final answer (2 s.f.).	**B1** **B1** **A1**

Question 2		
(a)(i)	T = 24 x 3600 = 86400 s	**A1**
(a)(ii)	Gravitational force provides centripetal force $\frac{GMm}{r^2} = mr\omega^2 = mr(\frac{2\pi}{T})^2$ $r^3 = \frac{GMT^2}{4\pi^2}$ $r = \sqrt[3]{\frac{(6.67\times10^{-11})(5.97\times10^{24})(86400)^2}{4\pi^2}} = 4.22\times10^7$ m	**B1** **A1**
(b)(i)	$\Delta\phi = -\frac{GM}{r_{far}} - (-\frac{GM}{r_{near}}) = GM(\frac{1}{r_{near}} - \frac{1}{r_{far}})$ $\Delta\phi = (6.67\times10^{-11})(5.97\times10^{24})(\frac{1}{6.87\times10^6} - \frac{1}{4.62\times10^7}) = 4.93\times10^7$ J kg^{-1}	**B1** **A1**
(b)(ii)	Loss in KE = Gain in GPE = $m\Delta\phi$ = $1500(4.93\times10^7) = 7.40\times10^{10}$ J Change in KE from closest to furthest = -7.40 x 10^{10} J Note to marker: Deduct 1 mark if a positive answer is given.	**M1** **A1**
(b)(iii)	The satellite moves slowest at its furthest point (due to minimum kinetic energy), meaning it spends the greatest proportion of its orbital time / lingers over the polar region.	**A1**

Question 9		
(a)	The **total linear momentum** of a system **remains unchanged** if **no net external force** acts on the system	**B1** **B1**
(b)(i)	Applying conservation of linear momentum: $m_A u_A + m_B u_B = m_A v_A + m_B v_B$ $(1.2)(6.40) + (5.2)(0.30) = 1.2v_A + 5.2v_B$ Since collision is elastic, relative speed of approach = relative speed of separation $u_A - u_B = v_B - v_A$ $v_A = v_B - 6.10$ Solving, $1.2v_A + 5.2(v_A + 6.10) = 9.24$ or $1.2(v_B - 6.10) + 5.2v_B = 9.24$ $v_B = 2.5875 = 2.6 \text{ m s}^{-1}$	**B1** **B1** **B1** **A0**
(b)(ii)	 From **(i):** $v_A = 2.6 - 6.1 = -3.5 \text{ m s}^{-1}$ • Correct flat lines for where relevant • A smooth and downward sloping curve in between One mark is deducted for any mistake made	**A1** **A2**
(b)(iii)	Draw a tangent at the time on the curve of variation of velocity with time of trolley B. Determine its gradient and multiply it with the mass of trolley B to arrive at the force on trolley B.	**B1** **B1**

(b)(iv)	The total kinetic energy decreases upon collision and reaches a (non-zero) minimum value. After which, total kinetic energy increases and returns to its initial total kinetic energy before collision.	**B1** **B1**
(b)(v)	At maximum compression, the two trolleys have the same speed. Applying conservation of linear momentum: $m_A u_A + m_B u_B = (m_A + m_B)v_C$ $(1.2)(6.40) + (5.2)(0.30) = (1.2 + 5.2)v_C$ $v_C = 1.444 \text{ m s}^{-1}$ Applying conservation of energy Gain in E.P.E. = Loss in K.E. $= (0.5)(1.2)(6.40^2) + (0.5)(5.2)(0.30^2) - (0.5)(1.2 + 5.2)v_c^2$ $= 18.14$ $= 18.1 \text{ J}$ EPE in each spring = 9.1 J	**B1** **B1** **A1**
(c)(i)	v_Y is the vertical velocity of trollley before colliding with the ground, $v_y^2 = 2(9.81)(10.0)$ $v = 14.0 \text{ ms}^{-1}$ Average vertical force = $= \frac{m\Delta v}{\Delta t} = \frac{5.2(14.0 - 0)}{0.100} = 728 \text{ N}$ (upwards) Applying Newton's second law to the vertical direction: $N_C - mg = 728$ $N_C = 728 + 5.2(9.81) = 779 \text{ N}$  Note to marker: Award max 2 marks for students who ignore weight in their analysis.	**B1** **B1** **B1** **A1**
(c)(ii)	Impulse is the change in momentum, so impulse (in the vertical direction) is larger. (*Impulse in the horizontal direction depends on the angle of rebound. Students can discuss this but as long as their deduction makes sense.*)	**B1**

Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions required for a body to be in equilibrium.

The sum of forces acting on the body in any direction must be zero.
(The resultant force on the body is zero / No resultant force)

The sum of the moments of the forces about any point must be zero.
(The resultant moment (or torque) on the body about any axis is zero. / No resultant torque)

[2]

(b) Block A of mass 1.7 kg is connected by a light inextensible string to a counterweight B of mass *M* via a system of two massless and frictionless pulleys P and Q. Block A is balanced on a smooth plane inclined at 45°.

Pulley Q is a double pulley made up of two fixed wheels of different radii attached to a common axle, with the radii of the smaller and larger wheels *r* and 3*r* respectively as shown in Fig. 1.1.

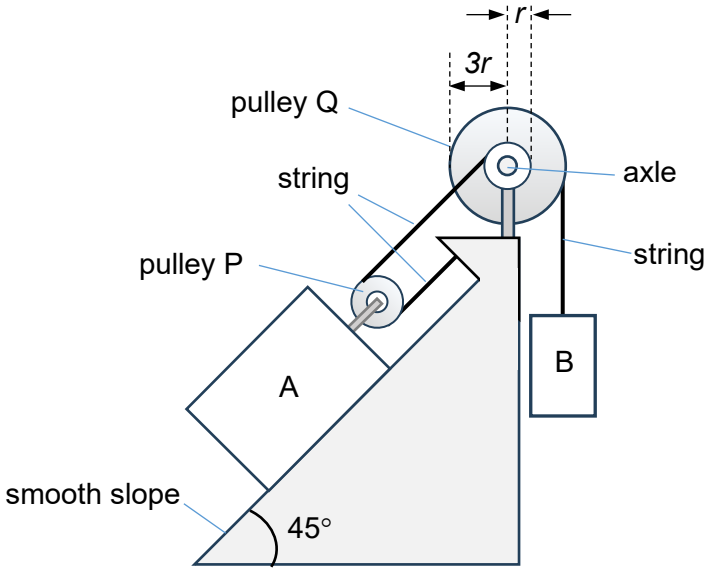
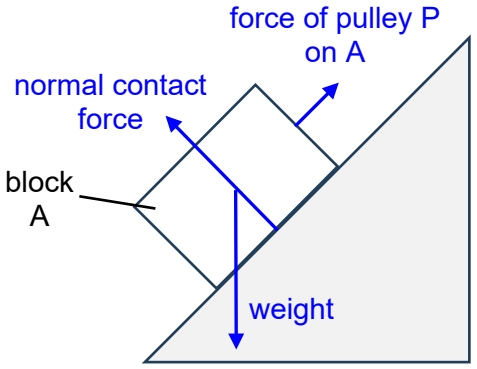


Fig.1.1

(i) Block A is in equilibrium. In Fig. 1.2, draw and label all forces acting on block A.



[B1] 3 correct forces
[B1] correct labels

Fig.1.2

[2]

(ii) Calculate the tension in the string connected to pulley P.

Block A is in equilibrium: $\sum F_{// \text{ slope}} = 0$

$m_A g \sin 45° = 2T$

$T = \frac{m_A g \sin 45°}{2} = \frac{(1.7)(9.81)\sin 45°}{2}$ [M1]

$= 5.90$ N [A1]

tension = 5.90 N N [2]

(iii) Determine the mass M of counterweight B.

Taking pivot about axel, $\sum \tau = 0$

CW moments = ACW moments

$m_B g(3r) = Tr$

$m_B = \frac{Tr}{g(3r)} = \frac{T}{3g}$ $\frac{5.90}{(3)(9.81)}$ [M1]

$= 0.200$ N [A1]

M = 0.200 kg kg [2]

(iv) Suggest and explain a change that can be made to the pulley such that block A will accelerate up the slope.

The larger wheel of the double pulley can be made larger with the smaller wheel remaining constant / more than 3 times larger than the smaller wheel. [M1]

This is so that the clockwise moment will be larger than the anticlockwise moment / net clockwise torque causing the block A to slide up the slope. [A1] [2]

[Total: 10]

2 A plane is flying with a velocity of 240 m s^{-1} at an angle of 30° with respect to the horizontal, as shown in Fig. 2.1. At an altitude of 2.0 km, a projectile is released from the plane. The projectile hits the target on the ground. Assume air resistance is negligible.

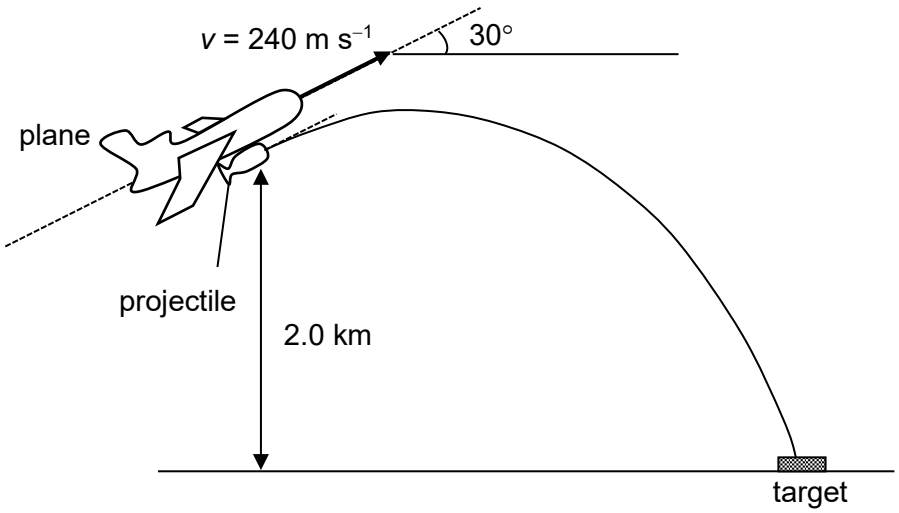


Fig. 2.1

(a) Determine the horizontal displacement of the projectile from the point of release to the target.

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$-2000 = (240 \sin 30)t + \frac{1}{2}(-9.81)t^2 \quad \text{[C1]}$$

$$4.905t^2 - 120t - 2000 = 0$$

$$t = -11.38 \text{ (reject)}, 35.8413 \text{ s}$$

$$s_x = u_x t$$

$$= (240 \cos 30)(35.8413) \quad \text{[C1]}$$

$$= 7400 \text{ m} \quad \text{[A1]}$$

horizontal displacement = ... m [3]

(b) Determine the speed of the projectile just before it hits the target.

$$v_y = u_y + a_y t$$

$$= (240 \sin 30) + (-9.81)(35.84)$$

$$= -231.6$$

$$v = \sqrt{v_x^2 + v_y^2}$$

$$= \sqrt{240 \cos 30^2 + 231.6^2} \quad \text{[C1]}$$

$$= 311.2 \approx 310 \text{ m s}^{-1} \quad \text{[A1]}$$

OR

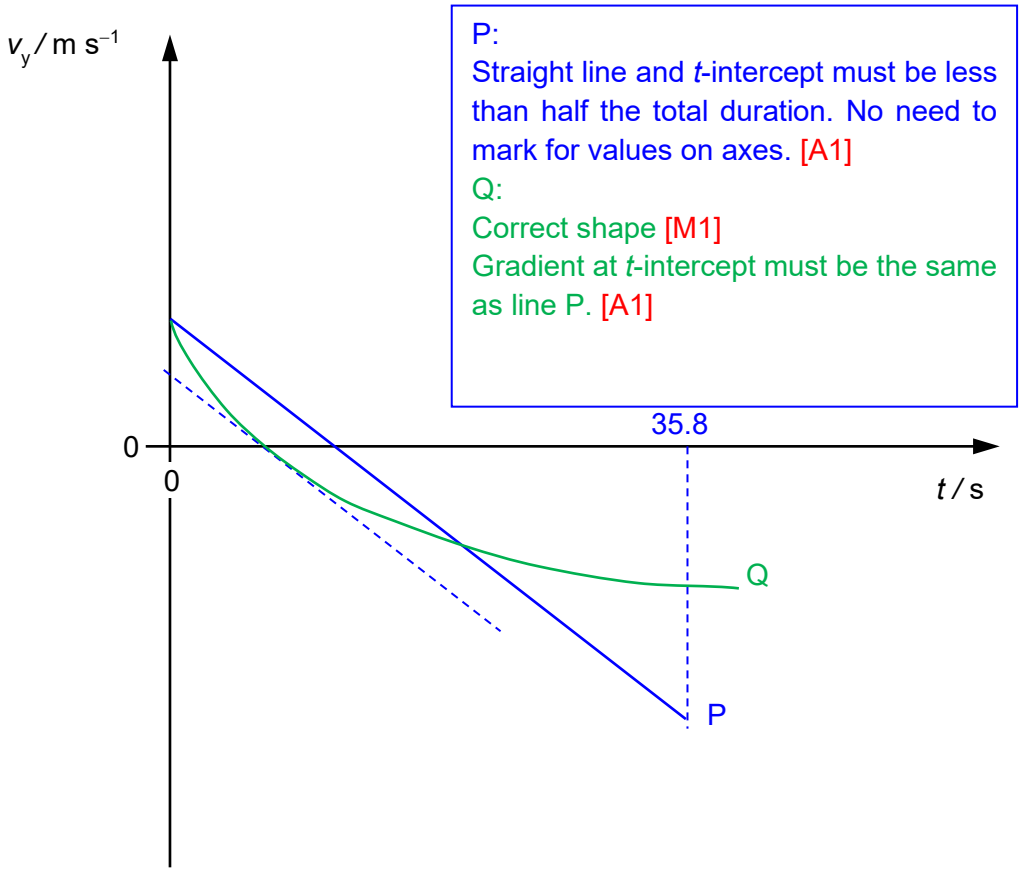
$$KE_{final} = KE_{initial} + GPE_{loss}$$

$$\frac{1}{2} m v_f^2 = \frac{1}{2} m (240)^2 + m(9.81)(2000) \quad \text{[C1]}$$

$$v_f = 311.2 \approx 310 \text{ m s}^{-1} \quad \text{[A1]}$$

speed = ... m s^{-1} [2]

(c) (i) Without further calculation, on Fig. 2.2, sketch the variation of vertical velocity v_y with time t of the projectile from the time it was released to the time it hits the target. Label this line **P**. Include values on your axes. [1]



P:
Straight line and t-intercept must be less than half the total duration. No need to mark for values on axes. [A1]
Q:
Correct shape [M1]
Gradient at t-intercept must be the same as line P. [A1]

Fig. 2.2

(ii) On Fig. 2.2, sketch the variation of vertical velocity v_y with time t of the projectile if air resistance was **not** negligible. Label this line **Q**. No calculations are required. [2]

[Total: 8]

3 **(a)** PQRS comprises of a rough slope PQR of height 1.1 m, and smooth horizontal surface RS. PQR of total surface length 2.7 m is made up of a straight section PQ and curved section QR as shown in Fig. 3.1.

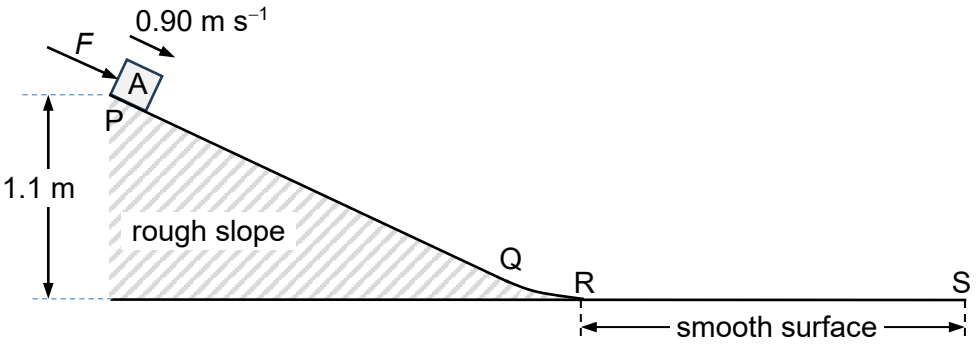


Fig. 3.1

Block A of mass 2.0 kg has an initial speed of 0.90 m s^{-1} at point P on rough slope. A force *F* is applied to block A as it moves along PQR. The magnitude of the *F* is 11 N and its direction is the same as the object's velocity at any instant along surface PQR.

The frictional force between the block A and the surface PQR may be taken to be constant at 6.4 N.

(i) Determine the speed of the block when it reaches R.

gain in KE = WD by gravitational force + WD by F − WD by friction

$= mgh + F.s - F_{friction}.s$

$= (2.0)(9.81)(1.1) + (11)(2.7) - (6.4)(2.7)$ [B1]

$= 34.002 = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$v = \sqrt{\frac{2(34.002)}{m} + u^2}$

$= \sqrt{\frac{(2)(34.002)}{2.0} + (0.90)^2}$ [B1]

$= 5.9$ m s^{-1} [A1]

speed = ... m s^{-1} [3]

(ii) Use Newton's second law to explain how the speed of block A varies along PQRS.

Along PQ the net force given by (F + component of weight parallel to slope $W_{//}$ – friction) = ma. (acceleration is constant and) Speed increases as $F_{net} > 0$ (as *F* > friction).

For the section QR, as $W_{//}$ decreases to zero as the slope flattens out, F_{net} is still > 0, speed of block increases further (at a decreasing rate).

(F_{net} goes from (F + component of weight parallel to slope $W_{//}$ – friction) to (F – friction)

For RS, the block moves at constant speed as F_{net} = 0, no applied force and no friction between surfaces.

[3]

(b) Block X of mass 0.27 kg moves with a speed 0.67 m s^{-1} on a smooth horizontal surface. Block Y of mass 0.34 kg moves with speed 0.46 m s^{-1} in the opposite direction. A light spring of natural length 13.0 cm and force constant 120 N m^{-1} is attached to block Y as shown in Fig. 3.2.

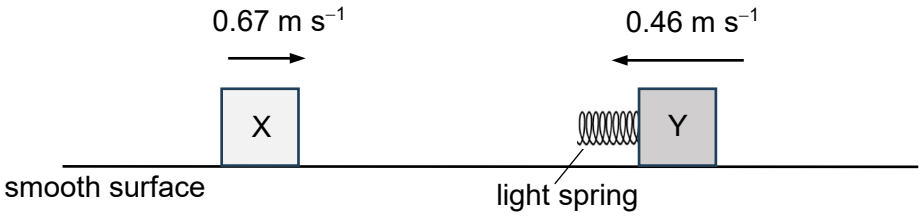


Fig. 3.2

When block X collides elastically head-on with block Y, there will be an instant when the spring experiences maximum compression.

Determine the closest distance of approach of the two blocks.

At closest distance of approach, both block moves with the same speed. [C1]
Taking rightwards as positive

$$m_A u_A + m_B u_B = (m_A + m_B)v$$

$$v = \frac{m_A u_A + m_B u_B}{m_A + m_B} = \frac{(0.27)(0.67) + (0.34)(-0.46)}{0.27 + 0.34} \qquad \text{[C1]}$$

$$= 0.0402 \text{ m s}^{-1}$$

Difference in KE before collision and max compression = EPE stored in spring

$$\left(\frac{1}{2}m_A u_A^{\,2} + \frac{1}{2}m_B u_B^{\,2}\right) - \frac{1}{2}(m_A + m_B)v^2 = \frac{1}{2}kx^2$$

$$\left(\frac{1}{2}(0.27)(0.67)^2 + \frac{1}{2}(0.34)(-0.46)^2\right) - \frac{1}{2}(0.27 + 0.34)(0.0402)^2 = \frac{1}{2}(120)x^2 \qquad \text{[C1]}$$

$$x = 0.0400 \text{ m}$$

$$\text{closest distance of approach} = L_0 - x$$

$$= 0.130 - 0.040 = 0.090 \text{ m} \qquad \text{[A1]}$$

closest distance of approach = ... m [4]

[Total: 10]

6 Read the passage and answer the questions that follow.

The Juno spacecraft is launched by NASA in 2011, and its mission is to study Jupiter's origin, interior structure, atmosphere, and magnetosphere to understand the solar system's formation by measuring composition, gravity, and magnetic fields with its myriad of sensors on board.

This is also NASA's first mission to Jupiter to use solar panels rather than radioisotope thermoelectric generators (RTGs) for powering the spacecraft. Jupiter orbits the Sun at a much greater distance than Earth, making solar power generation a significant engineering challenge.

Solar radiation spreads out as it travels from the Sun. The solar intensity I at a distance d from the Sun follows the inverse square law. The solar intensity on Jupiter is only 3.72 % that on Earth. To compensate for the low intensity at Jupiter, Juno is equipped with three large solar array wings, as shown in Fig 6.1.

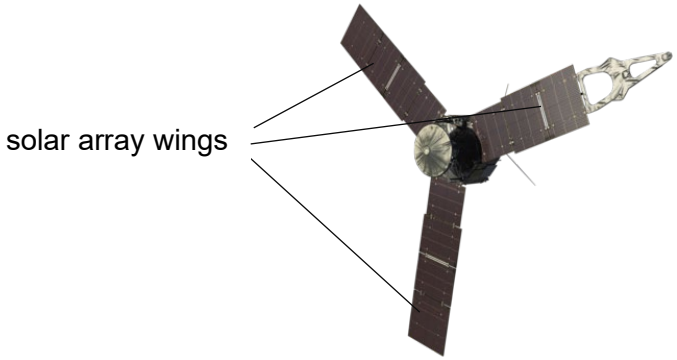


Fig. 6.1

The electrical power P generated by the solar panels on the wings is given by the equation:

$$P = I A \eta$$

where A is the total surface area of the solar panels and η is the efficiency of the solar cells. The efficiency of the cells is affected by the extremely low temperatures in the outer solar system.

Jupiter has an intense radiation belt above its equator that spans a distance of $2R_J$ to $10R_J$ from its centre, where R_J is the radius of Jupiter. Exposure to excessive radiation can cause damage to the spacecraft and its sensors. To minimize such exposure, Juno follows a highly elliptical polar (pole-to-pole) orbit, as shown in Fig 6.2.

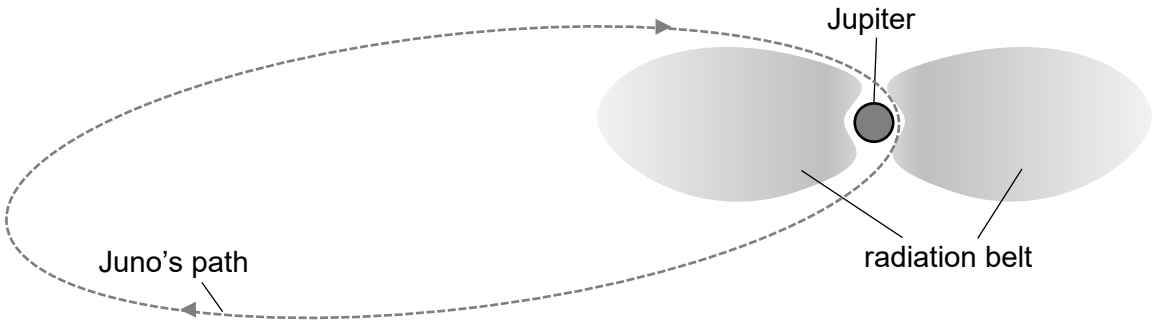


Fig. 6.2

Once Juno enters Jupiter's orbit, its propulsion system is turned off. It is subsequently used sparingly only to correct its path when necessary.

The distance of Juno from the centre of Jupiter in the first three months of 2017 is shown in Fig. 6.3.

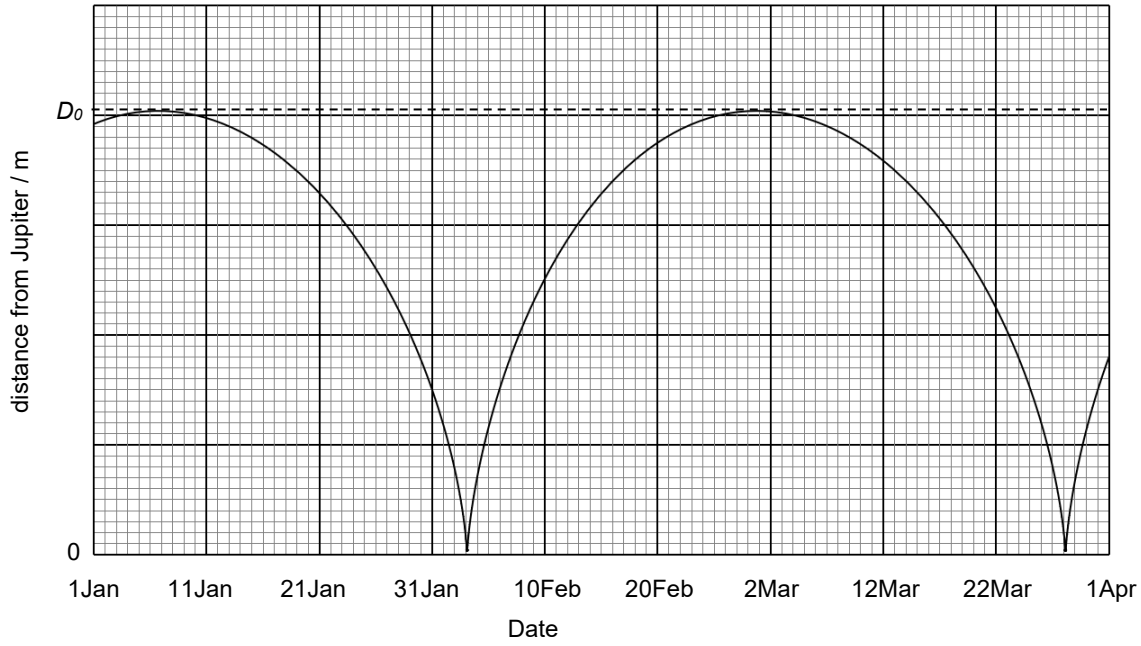


Fig. 6.3

Due to the vast distances involved, it took almost 5 years for Juno to enter Jupiter's orbit from Earth. Communication between Juno and Earth which is conducted via radio waves also experiences significant "light time" delay for signals.

Additional data for Earth, Jupiter and the Juno mission are provided in Table 6.4.

Property	**Value**
Mass of Jupiter M_J	1.90×10^{27} kg
Radius of Jupiter R_J	7.15×10^{7} m
Mean distance* of Earth from the Sun d_E	1.50×10^{11} m
Solar intensity at Earth's distance I_E	1360 W m^{-2}
Total surface area of Juno's solar panels A	60 m^2
Efficiency of solar cells at Jupiter η	24%
Mass of Juno spacecraft m	3625 kg
Perijove (closest distance* to Jupiter) r_p	1.06 R_J
Apojove (furthest distance* from Jupiter) r_a	113 R_J

*center to center

Table 6.4

(a) Explain why the solar intensity I follows an inverse square law as the distance d from the Sun increases.

Sun is taken as a point source, and the light radiates uniformly radially spherical surface area is proportional to the square of distance from the Sun [B1]. Since intensity is inversely proportional to area [B1], it is inversely proportional to the square of the distance.

[2]

(b) Using the relation in **(a)** and the data provided, show that the mean distance of Jupiter from the Sun is 7.78×10^{11} m.

$$\frac{\text{Intensity on Jupiter}}{\text{Intensity on Earth}} = 0.0372 = \left(\frac{\text{Distance of Earth}}{\text{Distance of Jupiter}}\right)^2 \quad \text{[B1]}$$

$$\text{Distance of Jupiter} = \frac{1.50 \times 10^{11}}{\sqrt{0.0372}} \text{ [B1]} = 7.78 \times 10^{11} \text{ m}$$

[2]

(c) Calculate the total electrical power output generated by Juno's solar panels when Jupiter is at its mean distance from the Sun.

$P = I A \eta = 50.6 \times 60 \times 0.24$

$= 729$ W [A1]

electrical power = ... W [1]

(d) Suggest **one** reason, other than distance from the Sun, why the electrical power output might fluctuate during the mission.

Facing direction of the solar wings relative to the Sun / Jupiter positioned between Juno and the Sun / Any other reasonable answer. [A1]

[1]

(e) Determine from Fig. 6.3 the period of Juno's elliptical orbit.

86 – 33 = 53 days [A1]

period = ... days [1]

(f) Determine the value of D_0 in Fig. 6.3.

D_0 = apojove = $113 \times 7.15 \times 10^7 = 8.08 \times 10^9$ m [A1]

D_0 = ... m [1]

(g) Explain why the sum of Juno's kinetic energy and gravitational potential energy remains constant when it is in orbit.

With the propulsion system turned off, the only force that acts on Juno is the gravitational force by Jupiter. Since there are no external forces [B1] on the system of Juno and Jupiter, total energy must remain constant. [1]

(h) Calculate the difference between the maximum and minimum values for the kinetic energy of Juno during an orbit.

Difference in max and min values of KE = Difference in max and min values of GPE [B1]

$$= -GM_J m_j \left(\frac{1}{r_a} - \frac{1}{r_p}\right)$$

$$= -6.67\times10^{-11}\times1.90\times10^{27}\times3625\times\left(\frac{1}{8.08\times10^{9}} - \frac{1}{1.06\times7.15\times10^{7}}\right) \text{ [B1]}$$

$$= 6.00\times10^{12} \text{ J [A1]}$$

difference = ... J [3]

(i) On Fig. 6.2, indicate point X at which Juno is moving with maximum speed and point Y at which it is moving with minimum speed. [1]

(j) Explain using Fig. 6.2 and your answer in **(i)** how the highly elliptical orbit minimizes the spacecraft's exposure to Jupiter's intense radiation belt.

Fig. 8.2 shows that Juno is near or within the radiation belt for only a small portion of its orbit. [B1] In addition, from **(i)**, its speed when it is near or within the radiation belt is the greatest, hence the time it spends in the region is further minimised. [B1]

[2]

(k) For a body in a circular orbit, gravitational field strength g is always equal to $\frac{v^2}{r}$, where v is its linear speed and r is the radius of the orbit. In Juno's case, assume g is equal to $\frac{v^2}{r}$ at its perijove and its apojove. Use this relationship to calculate Juno's minimum speed.

Speed is minimum at the apojove:

$$g = \frac{GM_J}{r_a^2} = \frac{6.67\times10^{-11}\times1.90\times10^{27}}{\left(8.08\times10^{9}\right)^2} \text{ [B1]} = 1.94\times10^{-3} \text{ N kg}^{-1}$$

Since $g = \frac{v^2}{r}$, $v = \sqrt{gr} = \sqrt{1.94\times10^{-3}\times8.08\times10^{9}}$ [B1] $= 3.96\times10^{3}$ m s^{-1} [A1]

minimum speed = ... m s^{-1} [3]

(l) Explain why the relationship $g = \frac{v^2}{r}$ does not hold in general for elliptical orbits.

Juno's distance from Jupiter is not constant. [B1]

When Juno is moving towards Jupiter, $g > v^2/r$. When Juno is moving away from Jupiter, $g < v^2/r$. [B1], OR Juno's velocity v is almost always not perpendicular to g, thus there is a component of v that is causing it to move closer or further away from Jupiter. [B1] [2]

(m) Using the mean distances of Jupiter and of Earth from the Sun, estimate a minimum value for the "light time" delay for signals to travel from Jupiter to Earth.

Minimum distance between Jupiter and Earth = $(7.78 - 1.50)\times10^{11} = 6.28\times10^{11}$ m [B1]

Minimum time taken = $\frac{6.28\times10^{11}}{3.0\times10^{8}}$ = 2100 s [A1]

delay = ... s [2]

[Total: 22]

End of Paper

Section A

Answer **all** the questions in the spaces provided.

1 Two spheres A and B of masses 3.7 kg and 2.7 kg respectively are attached to two light inextensible strings L_1 and L_2 each of length 0.50 m.

The combination of masses is made to swing in a vertical circle with both strings always in a straight line, as shown in Fig. 1.1.

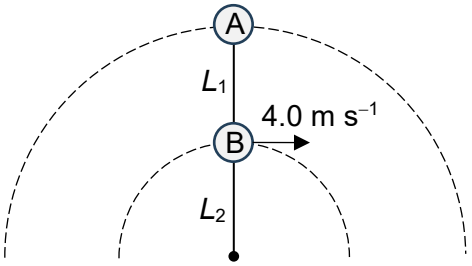


Fig 1.1

At the top of its motion, mass B is travelling at 4.0 m s^{-1}.

(a) Determine the speed of mass A at this instant.

Both masses have the same angular velocity.

$v = r\omega$

$\frac{v_A}{r_A} = \frac{v_B}{r_B} \Rightarrow \frac{v_A}{0.5+0.5} = \frac{4.0}{0.5}$ [M1]

$v_A = 8.0$ m s^{-1} [A1]

speed of mass A = ... m s^{-1} [2]

(b) Show that the tension in L_1 at this instant is 200 N.

$W_A + T_A = \frac{mv^2}{r}$

$T_A = \frac{mv^2}{r} - W_A$

$= \frac{(3.7)(8.0)^2}{0.50+0.50} - (3.7)(9.81)$ [M1]

$= 200.5 = 200$ N [A0]

[1]

(c) Hence or otherwise, determine the tension in L_2 at this instant.

$$W_B + T_B - T_A = \frac{m_B v_B^{\ 2}}{r_B}$$

$$T_B = \frac{m_B v_B^{\ 2}}{r_B} - W_B + T_A$$

$$= \frac{(2.7)(4.0)^2}{0.50} - (2.7)(9.81) + 201 \quad \text{[M1]}$$

$$= 261\ \text{N} \quad \text{[A1]}$$

tension in L_2 = ... N [2]

(d) At another instant in time, the combination of masses is at the bottom of its motion as shown in Fig. 1.2.

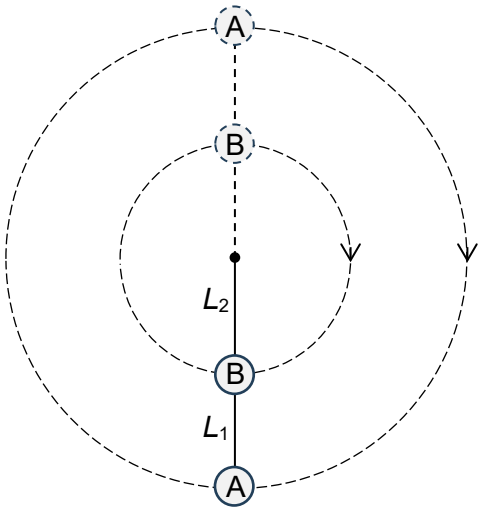


Fig 1.2

A student suggests that because the tensions in strings L_1 and L_2 change continuously during the motion, the total mechanical energy (i.e. kinetic and potential energy) of the system is not conserved.

With reference to tensions in the strings, state whether you agree with this statement and explain your reasoning.

I disagree with the statement, as tension is **always** acting perpendicular to the direction of motion of both masses [M1] Hence there is no work done by tension and hence total mechanical energy of the system is conserved. [A1]

I disagree with the statement, as tension is **always** acting perpendicular to the direction of motion of both masses. [M1] Hence there is no work done by tension but there may be other resistive forces and hence mechanical energy is not conserved. [A1]

[2]

[Total: 7]

2 **(a)** Define gravitational potential at a point

The gravitational potential at a point is defined as the work done per unit mass [B1]

by an external force in bringing a small test mass from infinity to that point. [B1]

[2]

(b) **(i)** The dwarf planet Ceres may be considered to be an isolated uniform sphere of mass 9.4×10^{20} kg and radius 4.7×10^{5} m.

A particle is launched vertically upwards from the surface of the planet. Assuming negligible resistance from the planet's atmosphere, determine the minimum speed of this particle that will result in it escaping from the gravitational pull of the planet Ceres.

By conservation of energy,

Total GPE + KE at surface = Total GPE + KE at infinite distance away [C1]

$$\text{GPE}_{surface} + \text{KE}_{surface} = \text{GPE}_{r=\infty} + \text{KE}_{r=\infty}$$

$$-\frac{GMm}{r} + \frac{1}{2}mv^2 = 0 + 0$$

$$\frac{GM}{r} = \frac{1}{2}v^2$$

$$\frac{(6.67 \times 10^{-11})(9.4 \times 10^{20})}{4.7 \times 10^{5}} = \frac{1}{2}v^2 \quad \text{[C1]}$$

$v = 520 \text{ m s}^{-1}$ [A1]

speed = ... m s^{-1} [3]

(ii) Hydrogen may be assumed to be an ideal gas.
The mass of a hydrogen molecule is 3.3×10^{-27} kg.

Calculate the root-mean-square (r.m.s.) speed of a hydrogen molecule in hydrogen gas that is at a temperature of 240 K.

$$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

$$\frac{1}{2}(3.3 \times 10^{-27}){c_{rms}}^2 = \frac{3}{2}(1.38 \times 10^{-23})(240) \quad \text{[C1]}$$

$$c_{rms} = 1700 \text{ m s}^{-1} \quad \text{[A1]}$$

r.m.s. speed = ... m s^{-1} [2]

(c) The surface of planet Ceres reaches temperatures of approximately 240 K when in direct sunlight.

Use your answers in **(b)(i)** and **(b)(ii)** to suggest a reason why Ceres does not have an atmosphere consisting of hydrogen.

Since r.m.s. speed of hydrogen molecules is more than the escape velocity on the surface of Ceres, the hydrogen molecules have sufficient kinetic energy to overcome the gravitational pull of the planet Ceres and escape to infinity. [1]

[Total: 8]

Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions for a system to be in static equilibrium.

Vector sum of forces acting on the system = 0 [1]

Sum of clockwise moments about a pivot = sum of anticlockwise moments about the same pivot. [1]

[2]

(b) A uniform wooden rod of length 0.30 m and mass 0.45 kg is attached to a vertical wall with a hinge at an angle of 40° with the horizontal. The other end of the rod is attached to the wall with a rope, so that the rope makes an angle of 30° with the horizontal. An object M of mass 1.2 kg is attached to the other end of the rod, as shown in Fig. 1.1 below.

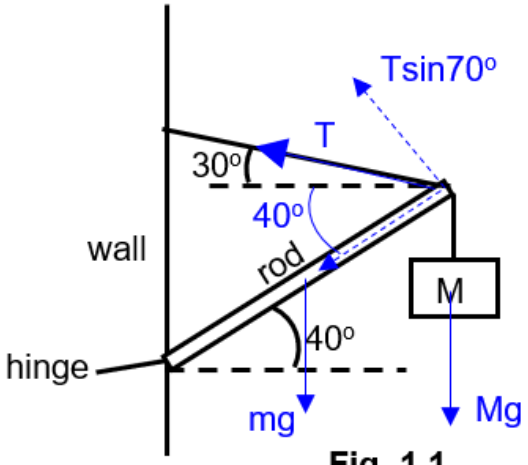


Fig. 1.1

(i) Show that the tension in the rope is 11 N.

Let length of rod be L.
Take moments about hinge.
Anticlockwise moment = clockwise moment

$$T\sin 70^\circ \times L = \left(Mg \times L\cos 40^\circ\right) + \left(mg \times \frac{L}{2}\cos 40^\circ\right)$$

$$T = \frac{\left(M + \frac{m}{2}\right) g\cos 40^\circ}{\sin 70^\circ}$$

$$= \frac{\left(1.2 + \frac{0.45}{2}\right) \times 9.81 \times \cos 40^\circ}{\sin 70^\circ}$$

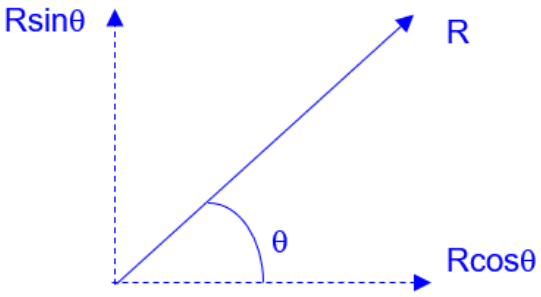
$$= 11.395$$

$$\approx 11\ \text{N}$$

[1 for correct anticlockwise moment; 1 for correct total clockwise moment
1 for more sf before final answer]

[3]

(ii) Determine the magnitude of the reaction force from the hinge acting on the rod, and the angle that the reaction force makes with the horizontal.



Take upwards and rightwards to be positive.

Let reaction force be R

Horizontally

$$R\cos\theta = T\cos 30^\circ$$
$$= 11\cos 30^\circ$$
$$= 9.526 \text{ N} \quad (1) \quad [1 \text{ mark}]$$

Vertically

$$R\sin\theta + T\sin 30^\circ = (M+m)g$$
$$R\sin\theta = (M+m)g - T\sin 30^\circ$$
$$= \left[(1.2+0.45)\times 9.81\right] - 11\sin 30^\circ$$
$$= 10.69 \text{ N} \quad (2) \quad [1 \text{ mark}]$$

$(2)\div(1)$ $\quad \tan\theta = \dfrac{10.69}{9.526}$

$$\theta = 48.30^\circ$$
$$\approx 48^\circ \quad [1 \text{ mark}]$$

Sub into (1) $\quad R\cos 48.30^\circ = 9.526$

$$R = 14 \text{ N} \quad [1 \text{ mark}]$$

magnitude = ... N

direction = ... ° from the horizontal

[4]

[Total: 9]

2 A speedboat of mass 450 kg is equipped with two identical motors. Each motor can produce 75 kW of power. When both motors are turned on, its maximum speed is 30 m s^{-1}. Assume that the drag force on the speedboat is proportional to the square of its speed.

(a) Determine the maximum speed of the boat when only one motor is turned on.

At maximum speed, driving force from motor = resistive force F.
Also $F = kv^2$, where k = constant.

Power supplied by motor $P = Fv_{max}$

$$= \left(kv_{max}^{\ 2}\right)v_{max}$$

$$= kv_{max}^{\ 3} \qquad [1]$$

$$\frac{P_1}{P_2} = \left(\frac{v_{max1}}{v_{max2}}\right)^3$$

$$v_{max1} = v_{max2} \times \sqrt[3]{\frac{P_1}{P_2}}$$

$$= 30 \times \sqrt[3]{0.5}$$

$$= 23.811$$

$$\approx 24 \text{ m s}^{-1} \qquad [1]$$

maximum speed = ... m s^{-1} [3]

(b) When both motors are turned on, the speedboat takes 9.0 s to reach the maximum speed of 30 m s^{-1}. Determine the amount of energy that is lost to the surroundings during that time.

Energy supplied by motor - energy lost = ke gained [1]

$$P_2 t - E_{lost} = \frac{mv^2}{2} - 0$$

$$E_{lost} = P_2 t - \frac{mv^2}{2}$$

$$= \left(150000 \times 9.0\right) - \frac{450 \times 30^2}{2}$$

$$= 1.1475 \times 10^6 \text{ J}$$

$$\approx 1.1 \times 10^6 \text{ J} \qquad [1]$$

amount of energy lost = ... J [2]

(c) Sketch, on Fig. 2.1, the variation with time t of the speed of the speedboat v, until it reaches maximum speed.

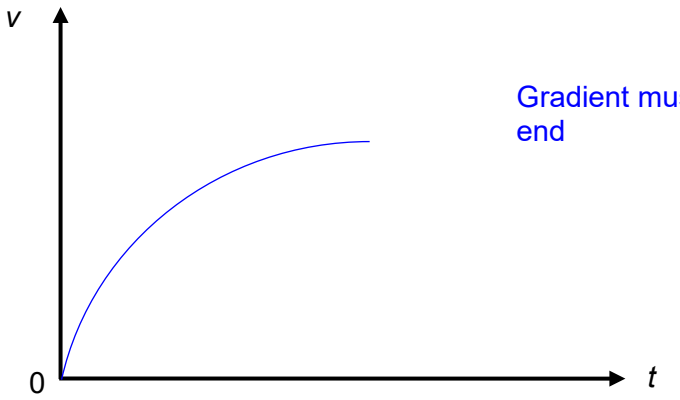


Fig. 2.1

[1]

(d) When the speedboat is travelling at a speed of 20 m s^{-1}, both motors are turned off at the same time. The speedboat continues to travel an additional distance of 600 m before coming to rest. Determine the average resistive force acting on the speedboat.

Work done against resistive force = ke lost [1]

$$fx = \frac{mv^2}{2} - 0$$

$$\text{Average resistive force } f = \frac{mv^2}{2x}$$

$$= \frac{450 \times 20^2}{2 \times 600}$$

$$= 150 \text{ N} \quad [1]$$

average resistive force = ... N [2]

[Total: 8]

3 **(a)** A small ball of mass 10 g rests at point P on a curved track of radius *r*, as shown in Fig. 3.1.

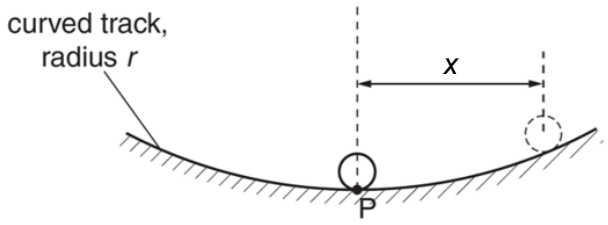


Fig. 3.1

The ball is moved a small distance to one side and is then released. The displacement *x* of the ball is related to its acceleration *a* towards P by the expression

$$a = -\frac{gx}{r}$$

where *g* is the acceleration of free fall.

(i) Explain how the above expression shows that the ball undergoes simple harmonic motion.

Since g/r is constant, a is proportional to x. [1]

The minus sign shows that a is opposite to x. [1]

[2]

(ii) The radius *r* of curvature of the track is 28 cm.
Determine the time interval τ between the ball passing point P and its first return to point P.

Compare given equation with $a = -\omega^2 x$,

$\omega^2 = \frac{g}{r}$ [1]

$(\frac{2\pi}{T})^2 = \frac{9.81}{28 \times 10^{-2}}$

T = 1.06 s [1]

Time taken between ball passing point P and its first return to point P

= $T/2$ = 1.06/2 = 0.53 s [1]

τ = ... s [3]

(b) The variation with time t of the displacement x of the ball in **(a)** is shown in Fig. 3.2.

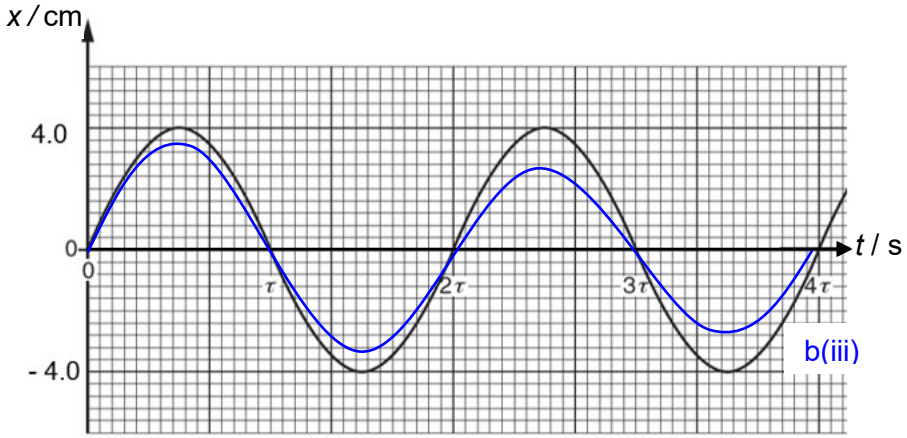


Fig. 3.2

(i) Calculate the maximum kinetic energy of the ball.

$$KE_{max} = \frac{1}{2} m\omega^2 x_o^{\ 2}$$

$$KE_{max} = \frac{1}{2}(10\times10^{-3})(\frac{9.81}{28\times10^{-2}})(4.0\times10^{-2})^2 \text{ [1]}$$

$$KE_{max} = 2.8\times10^{-4} \text{ J [1]}$$

maximum kinetic energy= .....J [2]

(ii) On Fig. 3.3, sketch the variation with time t of kinetic energy E of the ball for the first two periods. Include a suitable scale on the energy axis.

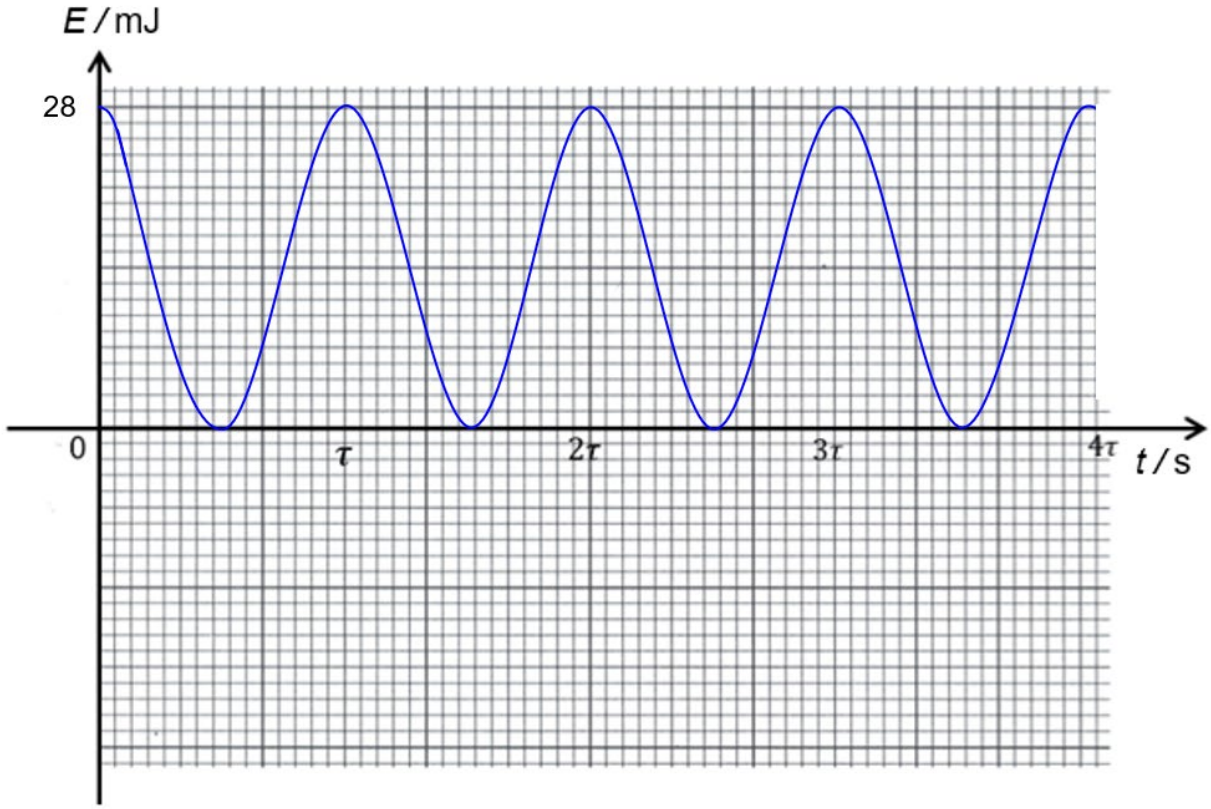


Fig. 3.3

[1]

(iii) Some moisture now forms on the track, causing the ball to come to rest after approximately after 15 oscillations.

On the axes of Fig. 3.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. [1]

Amplitude drops with time, period hardly changes.

[Total: 9]

7 Read the passage below and answer the questions that follow.

The Large Hadron Collider (LHC) is the world's largest and most powerful particle accelerator. It first started up on 10 September 2008 and is the crown jewel within the accelerator complex of the European Council for Nuclear Research (CERN).

Spanning across the Switzerland-France border, the LHC is a large horizontal ring in which high-energy particle beams, most commonly proton beams, are accelerated to speeds close to the speed of light in opposite directions. These proton beams circulate around the ring more than 11000 times a second as they are made to collide multiple times at different interaction points where the beams cross each other. By studying the fundamental particles produced in the aftermath of these proton-proton collisions, scientists hope to gain new insights and answers to some of the most important unsolved mysteries in Physics.

Inside the accelerator, the protons travel around the ring for hours along two separate beam pipes, both of which are kept at an ultrahigh vacuum. As the protons circulate around the LHC, they are repeatedly accelerated by strong electric fields along a short linear accelerator within the ring.

The protons are guided around the accelerator ring along the beam pipes by strong magnetic fields generated by thousands of electromagnets of different types and sizes, all of which need to be constantly kept at extremely low temperatures of about 1.9 K. This is why the LHC also houses the world's largest cryogenic system, using about 1.0×10^7 kg of liquid nitrogen and 1.3×10^5 kg of liquid helium to maintain the operation of the magnets.

The most common type of these electromagnets is called dipole magnets, which help bend the protons' trajectories so that they travel successfully along the curved beam pipes. These dipole magnets consist of many coils of electric cables placed next to and at many different sections along the beam pipe, each carrying a large electric current. The dipole magnets are contained and fastened together with strong non-magnetic stainless-steel collars.

Fig. 7.1 shows the cross-section of one of the dipole magnet setups at a section of a beam pipe of the LHC. The large number of electric cables are grouped into two double-layered regions around the beam pipe, labelled as region 1 and region 2 in Fig. 7.1. These electric cables are arranged parallel to the beam pipe such that when a current is travelling along these cables, the magnetic flux density produced by these cables at the centre of the beam pipe helps bend the protons' trajectories so that they can travel along the curved beam pipe.

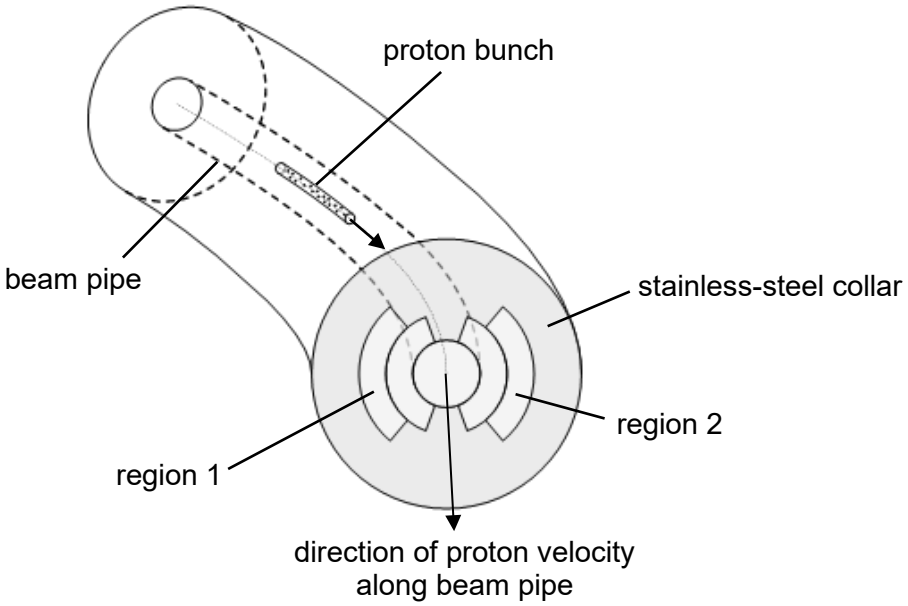


Fig. 7.1
(The electric cables parallel to the beam pipe are not shown for clarity.)

An important parameter in determining the functionality of a particle accelerator like the LHC is the beam current, which is defined as the average rate of flow of charge in the beam pipe. However, during normal operation, the protons do not travel in the beam pipe as a continuous beam. Instead, in each proton beam, the protons travel in groups called "bunches", which are approximately cylindrical in shape as shown in Fig. 7.1.

The LHC is built underground with a mean depth of 100 m for various reasons, including minimising the damage caused to the landscape and environment on the surface. In addition, protons travelling around the curved beam pipes accelerate to produce synchrotron radiation in the form of ultraviolet and X-ray photons, and interactions between the protons and the nuclei of any atoms around the beams in the beam pipes also produce ionising radiation. The Earth's crust therefore also provides natural shielding from these sources of radiation for anyone living on the surface near the LHC.

Table 7.1 shows some important data and parameters of the LHC under normal operating conditions.

Table 7.1

total circumference of LHC ring	26659 m
diameter of beam pipe	56 mm
proton energy	21.2 GeV
current in each cable of a dipole magnet setup	11850 A
length of a dipole magnet setup	14.3 m
peak dipole magnetic flux density at centre of beam pipe	8.33 T
beam current	0.58 A
number of bunches per proton beam	2808
length of one bunch	7.48 cm
cross-sectional area of one bunch	1.0 mm^2

(a) Suggest why the inside of the beam pipe needs to be maintained at an ultrahigh vacuum while the LHC is being operated.

This is to reduce collisions between protons and any gas molecules in the beam pipes, so that less harmful ionising radiation is produced.

OR [1]

This is to reduce collisions between protons and any gas molecules in the beam pipes, so that the proton beam can remain focused and not get scattered by these collisions.

OR

This is to reduce collisions between protons and any gas molecules in the beam pipes, so that there is less energy loss and the protons can reach the high speed required.

(b) State and explain if an accelerator like the LHC can be used to study neutron-neutron collisions.

No. As neutrons are neutral and uncharged, their trajectories cannot be bent / curved by the presence of magnetic fields.

OR [1]

No. As neutrons are neutral and uncharged, they cannot be accelerated by electric fields.

(c) (i) The protons in the LHC beam pipe travel at a speed of 0.999*c*, where *c* is the speed of light. **S**how that the period of revolution of the protons around the LHC ring is 8.89×10^{-5} s.

$$\text{Period } T = \frac{\text{distance travelled along the ring}}{\text{speed of protons}}$$
$$= \frac{26659}{0.999\,c}$$
$$= 8.8863 \times 10^{-5} \text{ s} \quad [1]$$
$$\approx 8.89 \times 10^{-5} \text{ s}$$

[1]

(ii) By considering the beam current, show that the total charge in the ring is 5.16 x 10^{-5} C.

Since beam current is defined as the average rate of flow of charge,

$$\text{beam current} = \frac{\text{total charge in the ring}}{\text{period}} = \frac{Q}{T}$$

$$0.58 = \frac{Q}{8.89 \times 10^{-5}}$$
$$Q = 5.156 \times 10^{-5} \text{ C} \quad [1]$$
$$\approx 5.16 \times 10^{-5} \text{ C}$$

[1]

(iii) Use your answer in **(c)(ii)** to determine the number of protons in each bunch within the proton beam.

Number of protons per proton beam,

$$N = \frac{Q}{e} = \frac{5.16 \times 10^{-5}}{1.60 \times 10^{-19}} = 3.225 \times 10^{14} \qquad [1]$$

Since there are 2808 bunches per proton beam,

$$\text{number of protons in each bunch} = \frac{3.225 \times 10^{14}}{2808}$$
$$= 1.15 \times 10^{11} \quad [1]$$

number of protons = ... [2]

(d) Fig. 7.2 shows a close-up of the cross-section of the dipole magnet setup around the beam pipe in Fig. 7.1, where regions 1 and 2 are represented by combined cables 1 and 2 respectively. Each combined cable carries electric current of the same magnitude.

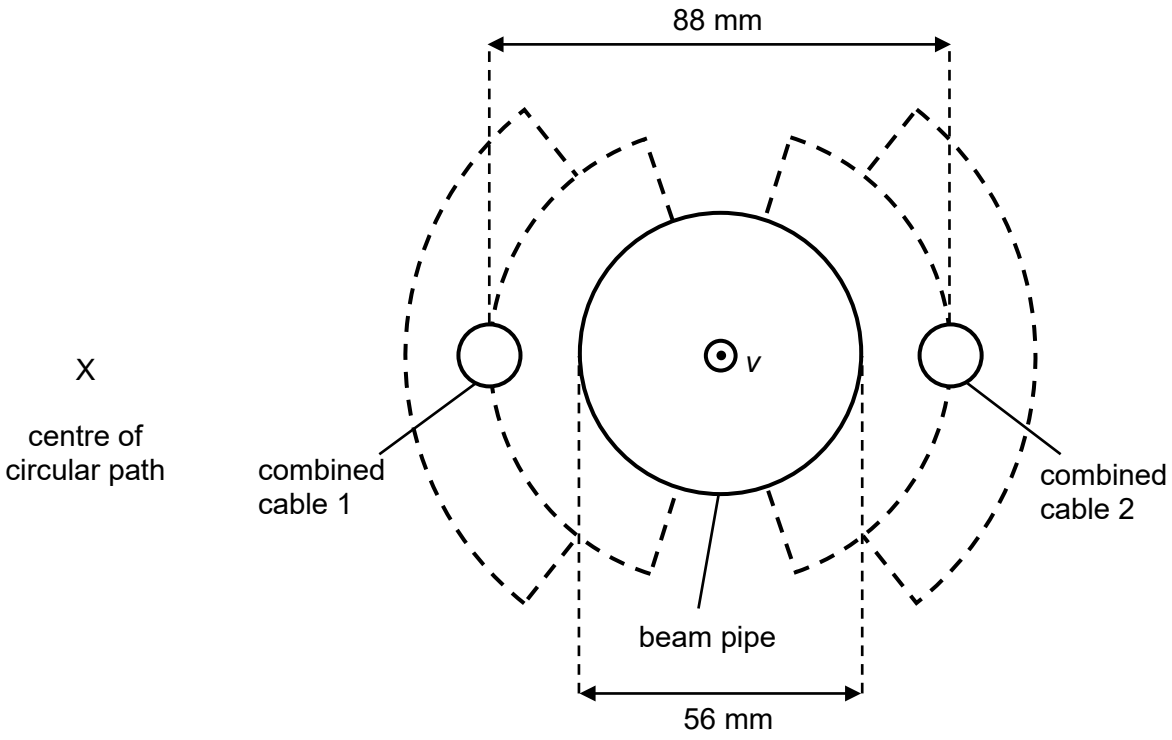


Fig. 7.2

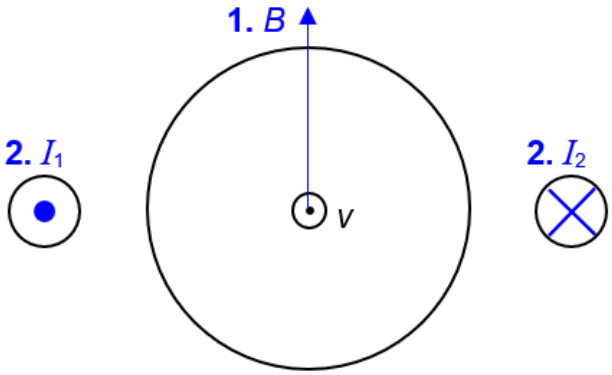
In this model of the dipole magnet setup, the distance between the combined cables is 88 mm, while the diameter of the beam pipe is 56 mm.

The proton velocity *v* in the middle of the beam pipe is pointing out of the page, as shown in Fig. 7.2. The centre of the circular path of the proton's trajectory is on the side of combined cable 1.

(i) On Fig. 7.2, using appropriate symbols and labels, indicate the direction of

1. the magnetic flux density *B* at the centre of the beam pipe,

2. the currents I_1 and I_2 in the combined cables 1 and 2 respectively. [2]



Since proton velocity is out of the page, beam current is out of the page.

Since centre of circular path of the proton's trajectory is on the left, the magnetic force on the proton that provides the centripetal force points to the left.

Using Fleming's left-hand rule, direction of *B* is upwards. [1]

*Straight line with arrowhead pointing upwards at the centre of pipe.

Since each combined cable carries current of the same magnitude, the direction of the current in each cable must be such that each produces a magnetic flux density that points upwards at the centre of the pipe, giving a resultant magnetic flux density as deduced in **(d)(i)1.**

Using the right-hand grip rule, I_1 points out of the page and I_2 points into the page. [1]

*Correct directions, positions and symbols of both currents.

(ii) 1. Determine the current required in each combined cable to produce the peak dipole magnetic flux density at the centre of the beam pipe.

$$B = \frac{\mu_0 I_1}{2\pi d} + \frac{\mu_0 I_2}{2\pi d}$$

$$8.33 = \frac{\left(4\pi \times 10^{-7}\right)\left(I_1 + I_2\right)}{2\pi\left(\frac{88}{2} \times 10^{-3}\right)}$$

$$I_1 + I_2 = 1832600 \text{ A} \quad [1]$$

Since $I_1 = I_2$,

$$I_1 = \frac{1832600}{2}$$

$$= 916300 \text{ A} = 9.16 \times 10^5 \text{ A} \quad [1]$$

current = ... A [2]

2. Hence, determine the number of cables in each region.

Since current in each cable is 11850 A,

$916300 = N(11850)$ where N is the number of cables

$N = 77.3 \approx 77$

number of cables = ... [1]

3. Suggest a reason why the dipole magnets need to be contained and fastened together with collars that are strong.

Support your reason with appropriate calculations.

Force that the combined cables exert on each other,

$$F = B_1 I_2 L$$

$$= \frac{\mu_0 I_1}{2\pi d} I_2 L$$

$$= \frac{\left(4\pi \times 10^{-7}\right)(916300)}{2\pi\left(88 \times 10^{-3}\right)}(916300)(14.3)$$

$$= 2.7287 \times 10^7 = 2.73 \times 10^7 \text{ N} \quad [1]$$

The cables in the dipole magnet setup experience an extremely large repulsive force away from each other. [1] Hence, strong collars are needed to keep the cables from moving away from the beam pipe. [1] .....[3]

(e) Besides dipole magnets, another type of electromagnet used at the LHC are quadrupole magnets.

Quadrupole magnets help "squeeze" the travelling protons within each bunch closer together so that the protons stay travelling along the central axis of the beam pipe in a tightly focused beam. They do so by accounting for the electrostatic repulsion between the protons within each bunch.

(i) By approximating the average volume occupied by a proton in each bunch to be the volume of a cube, show that the average distance between two adjacent protons in a bunch is 8.66×10^{-7} m.

Average volume occupied by one proton

$$= \frac{\text{volume of one bunch}}{\text{no. of protons in one bunch}}$$
$$= \frac{\left(7.48\times10^{-2}\right)\left(1.0\times10^{-6}\right)}{1.15\times10^{11}}$$
$$= 6.5043\times10^{-19}\ \text{m}^3 \qquad [1]$$

Approximating volume occupied by a proton to be volume of a cube with sides of length d:

$$d = \sqrt[3]{6.5043\times10^{-19}}$$
$$= 8.6643\times10^{-7}\ \text{m} \qquad [1]$$
$$\approx 8.66\times10^{-7}\ \text{m} \qquad [2]$$

(ii) Hence determine the magnitude of the average repulsive force between two adjacent protons in a bunch.

Average electric force of repulsion between 2 protons,

$$F_E = \frac{e^2}{4\pi\varepsilon_0 d^2}$$
$$= \frac{\left(1.60\times10^{-19}\right)^2}{4\pi\left(8.85\times10^{-12}\right)\left(8.6643\times10^{-7}\right)^2}$$
$$= 3.0663\times10^{-16} = 3.07\times10^{-16}\ \text{N} \qquad [1]$$

average repulsive force = ... N [1]

(iii) The quadrupole magnets also help to keep the protons in the proton beam on their intended path by accounting for the protons falling downwards due to their weight.

Determine the number of rounds a proton will be able to travel around the LHC accelerator ring before falling to the bottom of the beam pipe due to gravity, if there were no quadrupole magnets.

Assume that the acceleration due to gravity in the pipe is 9.81 m s^{-2}.

Initial velocity v of proton is horizontal along the central axis of the beam pipe.

Displacement of proton from centre of beam pipe to bottom of beam pipe,

$$s_y = u_y t + \frac{1}{2} g t^2$$

$$\frac{1}{2}\left(56 \times 10^{-3}\right) = 0 + \frac{1}{2}(9.81) t^2$$

$t = 0.075554$ s or -0.075554 s (NA) [1]

Total horizontal displacement travelled by proton,

$$s_x = vt$$
$$= (0.999)\left(3.00 \times 10^8\right)(0.075554)$$
$$= 22666199.8 \text{ m} \quad [1]$$

$$\text{Number of rounds} = \frac{s_x}{\text{circumference of ring}}$$
$$= \frac{22666199.8}{26659}$$
$$= 850.23 = 850 \quad [1]$$

number of rounds = ... [3]

[Total: 20]

Section A

Answer **all** the questions in the spaces provided.

1 A ball is thrown horizontally from the top of a building, as shown in Fig. 1.1.

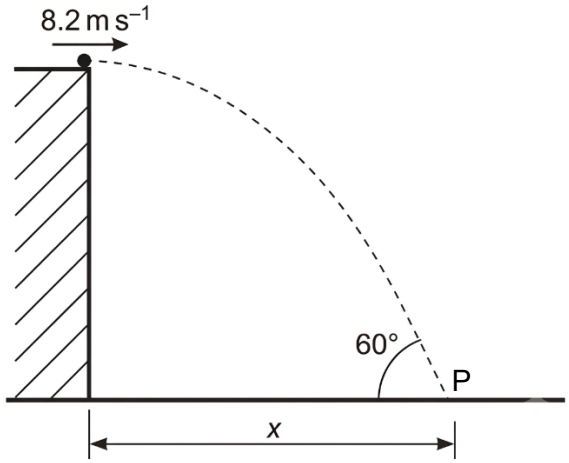


Fig. 1.1

The ball is thrown with a horizontal speed of 8.2 m s^{-1}. The side of the building is vertical. The ball hits the ground at point P at a distance *x* from the building and at an angle of 60° to the horizontal. Air resistance is negligible.

(a) For the ball at point P,

(i) show that the vertical component of its velocity is 14 m s^{-1},

$$\tan\theta = \frac{v_y}{v_x} \qquad [1]$$

$$v_y = v_x \tan\theta$$
$$= 8.2\tan 60°$$
$$= 14.2 \quad [1]$$
$$= 14 \text{ m s}^{-1}$$

[2]

(ii) determine the vertical distance it has fallen,

$$v_y^2 = u_y^2 + 2a_y s_y$$
$$14^2 = 0 + 2(9.81)s_y \quad \text{(Take downwards as +ve)} \qquad [1]$$
$$s_y = 9.99$$
$$= 10 \text{ m (2 sf)} \qquad [1]$$

vertical distance = ... m [2]

(iii) determine the horizontal distance *x*.

Let t be the time taken to fall through the height:

$v_y = u_y + a_y t$

$14 = 0 + 9.81t$

$t = 1.43$ s [1]

Then $s_x = u_x t$

$= 8.2 \times 1.43$

$= 11.7$

$= 12$ m (2 sf) [1]

distance *x* = ... m [2]

(b) State and explain the effect on your answer to the horizontal distance *x* if air resistance is not negligible.

Air drag causes the horizontal velocity to continually decrease throughout the motion. [1]

Because the average horizontal speed during the ball's flight is lower, the ball covers less horizontal distance in a given time [1] [2]

[Total: 8]

2 **(a)** State Newton's second law of motion.

The rate of change of momentum of an object is directly proportional to the resultant force acting on it, and takes place in the direction of the force. [1]

(b) A ball of mass 150 g travelling horizontally at a speed of 20 m s^{-1} collides with a vertical wall. Fig. 2.1 shows how the force of impact on the wall, *F*, varies with time *t*.

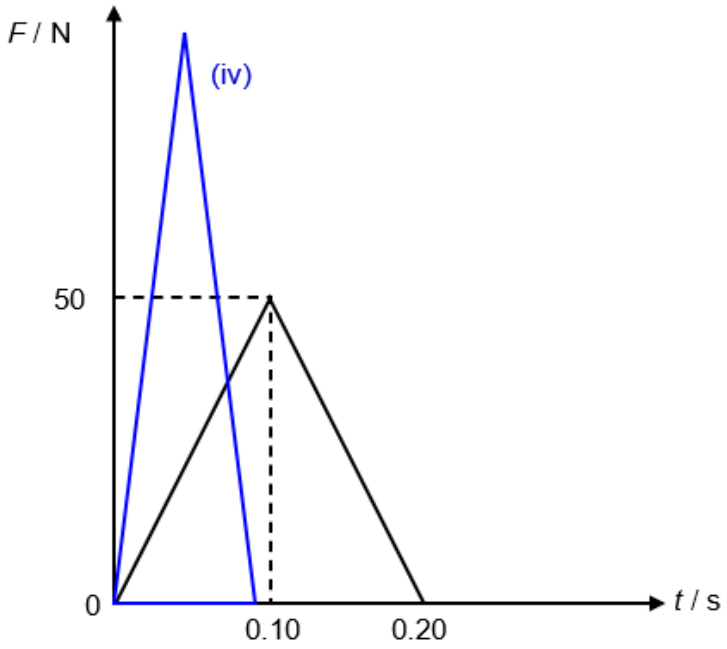


Fig. 2.1

(i) Determine the change in momentum of the ball.

Change in momentum = area under graph [1]

$$= \frac{50 \times 0.20}{2}$$

= 5.0 N s [1]

change in momentum = ... N s [2]

(ii) Hence determine the final speed of the ball.

Take rightwards as positive.
(Note: the impact force on the ball is to the left. So change in momentum of the ball is also to the left i.e. negative.)

Change in momentum = $m(v_f - v_i)$

$$-5.0 = 0.150(v_f - 20)$$

$$v_f = -13.33 \text{ m s}^{-1}$$

$\therefore$ Speed = 13 m s^{-1}

final speed = ... m s^{-1} [1]

(iii) Balls which are identical to the one above are thrown at the wall at the rate of 30 balls per minute. Determine the average impact force on the wall.

$$\text{Average force} = \frac{\text{Total change in momentum of 30 balls}}{\text{time interval}} \quad [1]$$

$$= \frac{30 \times 5.0}{60}$$

$$= 2.5 \text{ N} \quad [2]$$

average impact force = ... N [2]

(iv) A ball which is equal in mass as the one above is made of a harder material. It also collides with the wall at the same speed as before. Draw on Fig. 2.1 a second graph to show how the impact force on the wall varies with time when this ball collides with the wall. [1]

Higher peak and shorter time interval [1]

[Total: 7]

Section B

Answer **one** question from this section in the spaces provided.

8 A theme-park ride consists of two cages. They are moving in a circular path at constant speed v about a horizontal axis. Fig. 8.1 shows the ride at one instant when cage A is vertically above cage B.

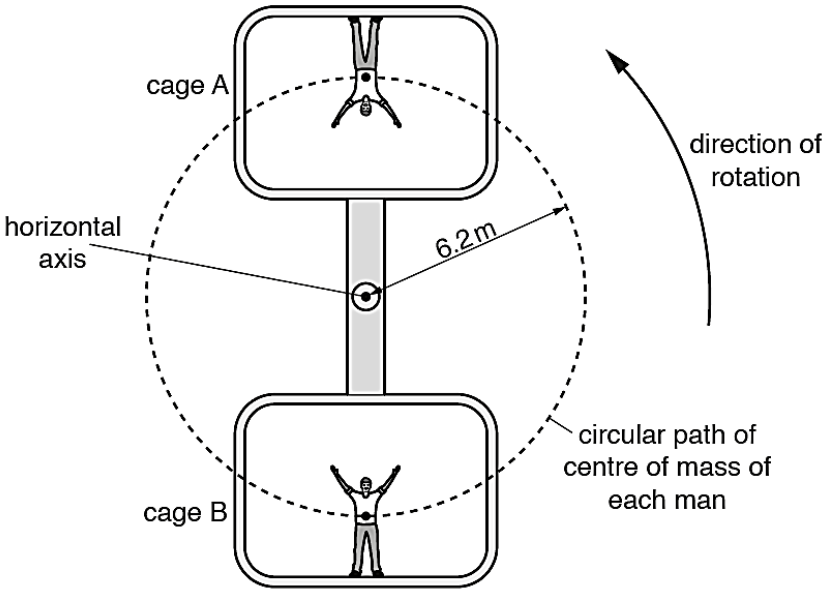


Fig. 8.1

(a) A man is riding in each cage. The mass of each man is 75 kg. The centre of mass of each man is 6.2 m from the horizontal axis. The period of one rotation is 4.1 s.

(i) Determine the speed v of the centre of mass of each man.

$$v = r\omega$$
$$= r\left(\frac{2\pi}{T}\right)$$
$$= 6.2\left(\frac{2\pi}{4.1}\right) \quad [1]$$
$$= 9.5 \text{ m s}^{-1} \quad [1]$$

speed v = ... m s^{-1} [2]

(ii) Calculate the magnitude of the acceleration of the centre of mass of each man.

$$a = r\omega^2$$
$$= r\left(\frac{2\pi}{T}\right)^2$$
$$= 6.2\left(\frac{2\pi}{4.1}\right)^2 \quad [1]$$
$$= 14.6 \text{ m s}^{-2} \quad [1]$$

acceleration = ... m s^{-2} [2]

(b) Fig. 8.2 shows the forces acting on the man in cage B at the instant the cages are in the positions shown. It shows the man in cage A at that same instant.

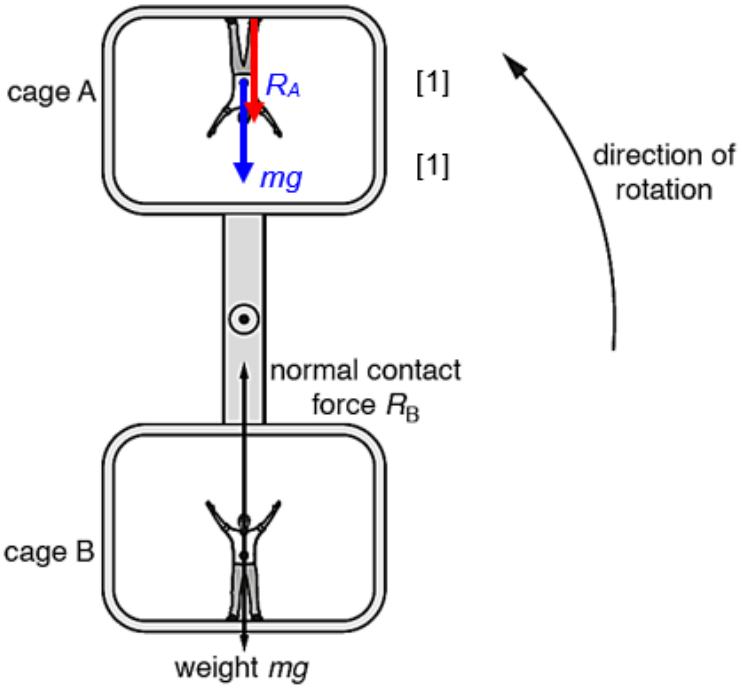


Fig. 8.2

(i) On Fig. 8.2, draw labelled arrows to represent the magnitude and direction of the forces acting on the man in cage A. [2]

(ii) Calculate the magnitude of the normal contact force R_A on the man in cage A at this instant.

$mg + R_A = ma_c$ [1]

$R_A = ma_c - mg$

$= 75(14.6 - 9.81)$

$= 360$ N (2 sf) [1]

R_A = ... N [2]

(c) **(i)** Explain why a minimum value for the speed is needed for the man in cage A to maintain in contact with the floor of his cage.

$R = ma - mg$, $R > 0$ is required to remain contact. [1]

so $a > g$, i.e. $v^2 / r > g$

$v > \sqrt{rg}$, the minimum speed for a contact force to be required [1] [2]

(ii) Calculate this minimum speed.

From (i), $v_{min} = \sqrt{rg}$

$= \sqrt{6.2 \times 9.81}$

$= 7.8$ m s^{-1} [1]

minimum speed = ... m s^{-1} [1]

(d) (i) State Newton's Law of Gravitation.

NLG states that two point masses attract each other with a force directly proportional to the product of the masses [1], and inversely proportional to the square of the distance between them. [1] [2]

(ii) The astronomer Johannes Kepler showed that the period T of rotation of a planet about the Sun is related to its mean distance R from the centre of the Sun by the expression

$$\frac{R^3}{T^2} = k$$

where k is a constant.

Show that, for planets in circular orbits about the Sun of mass M, the constant k is given by

$$k = \frac{GM}{4\pi^2}$$

where G is the gravitation constant.

Gravitational force provides the centripetal force [1]

$$\frac{GMm}{R^2} = mR\omega^2 \quad [1]$$

$$\frac{GMm}{R^2} = mR\left(\frac{2\pi}{T}\right)^2 \quad [1]$$

$$\frac{R^3}{T^2} = \frac{GM}{4\pi^2}$$

[3]

(iii) Earth can be assumed to be in a circular orbit about the Sun with an orbital radius of 1.50×10^8 km.

Use the expression in **d(ii)** to calculate a value for the mass of the Sun.

$$\frac{R^3}{T^2} = \frac{GM}{4\pi^2}$$

$$\frac{(1.50\times10^{11})^3}{(365\times24\times60\times60)^2} = \frac{6.67\times10^{-11}\times M}{4\pi^2} \quad [1]$$

$$M = 2.01\times10^{30} \text{ kg} \quad [1]$$

mass of Sun = ... kg [2]

(iv) Explain why, although planets experience gravitational forces from the Sun, they do not fall into the Sun.

At any instant, each planet is moving in a direction tangential to its orbit.[1] According to Newton's first law, the planet wants to continue to move in this direction [1] instead of falling into the Sun.

[2]

[Total: 20]